Schnorr Signatures are Tightly Secure in the ROM under a Non-Interactive Assumption

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Abstract. We show that the widely-used Schnorr signature scheme meets existential unforgeability under chosen-message attack (EUF-CMA) in the random oracle model (ROM) if the circular discrete-logarithm (CDL) assumption holds in the underlying group. CDL is a new, non-interactive and falsifiable variant of the discrete-logarithm (DL) assumption that we introduce. Our reduction is *completely tight*, meaning the constructed adversary against CDL has essentially the same running time and success probability as the assumed forger. This serves to justify the size of the underlying group for Schnorr signatures used in practice.

To our knowledge, we are the first to exhibit such a reduction. Indeed, prior work required interactive and non-falsifiable assumptions (Bellare and Dai, INDOCRYPT 2020) or additional idealized models beyond the ROM like the algebraic group model (Fuchsbauer, Plouviez and Seurin, EUROCRYPT 2020). To further demonstrate the applicability of CDL, we show that Sparkle+ (Crites, Komlo and Maller, CRYPTO 2023), a threshold signing scheme for Schnorr, is tightly secure (under static corruptions) assuming CDL. Finally, we justify CDL by showing it holds in two carefully chosen idealized models that idealize different aspects of the assumption.

Keywords: Schnorr signatures · threshold signatures · tight security · ECDSA conversion function

1 Introduction

1.1 Background and Main Results

<u>Schnorr signatures and our focus.</u> The Schnorr signature scheme [Sch90] (recalled below), specifically in the form of EdDSA [BDL$^+$12] implemented over twisted Edwards curves, is one of the most widely used pieces of cryptography today. For example, it is used in TLS, SSH, and in Bitcoin since the Taproot soft-fork upgrade in November 2021 [WNR20]. There is a rich theory behind the scheme's security, with tantalizing open questions. The initial result of Pointcheval and Stern (PS) [PS96] showed the scheme meets existential unforgeability (EUF-CMA) in the random oracle model (ROM) [BR93] assuming

the discrete-logarithm (DL) assumption holds in the underlying group. However, the PS result has the two major downsides, which have persisted despite much follow-on work: (1) the proof relies on the ROM, and (2) the reduction given in the proof is *lossy*. In this work, we focus on overcoming (2).

THE PROBLEM OF TIGHT SECURITY. Given a forger against the Schnorr signature scheme in the ROM with success probability p_{succ}, the PS result says there is an adversary solving DL in the underlying group with similar running time and having success probability $\Theta(p_{\mathsf{succ}}^2/q_H)$, where q_H is the number of RO queries made by the forger. Unfortunately, the discrepancy between the success probabilities of the assumed forger and the constructed DL adversary makes the result meaningless in practice. For example, suppose we implement the scheme over twisted Edwards curves of 256-bit order, which are conjectured to have 128-bit security for DL. Conservatively assuming at most 2^{64} RO queries, the PS result then tells us the scheme has $128/2 - 64 = 0$ bits of security for EUF-CMA! This is despite the lack of any known attack on the scheme short of solving DL after several decades of cryptanalysis.

So, can we give a tighter proof? Prior work [FPS20] showed tight security of Schnorr using additional idealized models such as the ROM combined with the algebraic group model (AGM) [FKL18], which makes assumptions on the adversary's strategy. Other work [NSW09, Sho23, CLMQ21] proved Schnorr secure directly in the generic group model (GGM) [Nec94, Sho97], making specific assumptions on the hash function. However, such idealized models are arguably better suited for analyzing simpler *assumptions* rather than the scheme itself.[3]

On the other hand, another sequence of works [PV05, GBL08, Seu12, FJS19] culminated in showing that there is *no* completely tight "generic" reduction in the ROM (even under a minimal formulation of security) for Schnorr signatures under *any* "representation-independent" non-interactive assumption. Later, Bellare and Dai (BD) [BD20] showed a generic reduction that loses *only* a q_H factor but still falls short of a completely tight reduction. (A subsequent result of Segev and Rotem [RS21] is less tight than BD's but uses a non-interactive assumption.)

This gap matters: going back to our previous example, BD guarantee $128 - 64 = 64$ bits of security for the scheme, which is insufficient for practical applications. BD also rely on a new *interactive* assumption they call multi-base DL, which is a variant of the one-more discrete logarithm assumption (OMDL) [BNPS03, BFP21]. Since the challenger must answer discrete-log queries by the adversary, multi-base DL is not a falsifiable assumption [Nao03].

We ask whether we can eliminate *both* downsides of their result, namely:

> *Is there a* completely tight *reduction in the ROM proving EUF-CMA of Schnorr signatures from a* non-interactive *and* falsifiable *assumption?*

Due to the above-mentioned impossibility results, such an assumption has to be *representation-dependent*, meaning it depends on the specific group representa-

[3] For example, these idealized models are subject to *uninstantiability* results [Den02, Zha22], so must be used with caution. Complex and interactive problems shown to hold in these models are more likely to fall prey to the reach of these results.

tion (see further discussion below). It is *a priori* unclear (to us, at least) what such an assumption would look like.

<u>Our results.</u> We resolve the above question in the affirmative. To do so, our central conceptual contribution is the *circular discrete-logarithm* (CDL) problem, a new, non-interactive, falsifiable, and representation-dependent variant of DL. We show that if CDL holds then the Schnorr signature scheme is secure in the underlying group, with a *completely tight* reduction in the ROM. While our assumption inherits the peculiar structure of Schnorr, we stress that our results improve on BD in two directions, firstly by using a *non-interactive and falsifiable* assumption and secondly by giving a *completely tight reduction*. We show further applicability of CDL by giving a tight proof of security for the recent three-round threshold-signing scheme Sparkle+ [CKM23b, CKM23a] for Schnorr signatures. (Their full version [CKM23a] corrects an error in the proceedings version.) This is relevant as Sparkle+ has seen standardization efforts [BD24].

Using carefully chosen idealized models that idealize different aspect of our assumption in appropriate elliptic-curve groups (*e.g.*, twisted Edwards curves), we show that breaking CDL has essentially the same complexity as breaking DL. Consequently, if one believes in these models for analyzing DL/CDL in appropriate elliptic-curve groups, then the security level of Schnorr in such groups shown by our results exactly matches that indicated by decades of cryptanalysis. Our results similarly establish that there is unlikely any faster attack on Sparkle+ than just solving DL.

1.2 Technical Overview

<u>The Schnorr signature scheme.</u> The Schnorr signature scheme is defined over a group $\mathbb{G}$ of prime order p generated by $g \in \mathbb{G}$ and uses a hash function $H\colon \mathbb{G} \times \{0,1\}^* \to \mathbb{Z}_p$. A secret key x is uniformly sampled from $\mathbb{Z}_p$, *i.e.*, $x \leftarrow\!\!\$\, \mathbb{Z}_p$, and defines a public key $h \leftarrow g^x$. A signature on a message $m \in \{0,1\}^*$ is computed by sampling $r \leftarrow\!\!\$\, \mathbb{Z}_p$, setting $R \leftarrow g^r$, computing $c \leftarrow H(R,m)$ and returning (R,s) with $s \leftarrow (r + cx) \bmod p$. A signature (R,s) on m under public key h is verified by checking $g^s = R \cdot h^c$, where $c \leftarrow H(R,m)$. We denote the scheme by $\mathsf{Sch}[\mathbb{G}, H]$ and usually drop H when working in the ROM.

<u>The circular discrete-logarithm problem.</u> Let $\mathbb{G}$ be as above, and let $f\colon \mathbb{G} \to \mathbb{Z}_p$ be an efficient function, which we call a *conversion function* following the terminology regarding ECDSA [Nat13]. We say that the *circular discrete-logarithm* (CDL) problem holds in $\mathbb{G}$ for f if, given uniformly sampled $h \in \mathbb{G}$, it is hard to find (R,z) such that

$$g^z = R \cdot h^{f(R)} \tag{1}$$

and $f(R) \neq 0$. One can think of f as a very simple function, not (necessarily) a cryptographic hash function used to instantiate Schnorr. We say that CDL holds for $\mathbb{G}$ if there exists f such that CDL holds in $\mathbb{G}$ for f. (We exclude the all-zeros function, since CDL trivially holds for it, as there is no solution.) We denote CDL for f in $\mathbb{G}$ as CDL[$\mathbb{G}, f$] and just CDL for $\mathbb{G}$ as CDL[$\mathbb{G}$]. The "circularity" in CDL is that in the solution equation R occurs both in the base and the exponent, mirroring this peculiar property of Schnorr's verification equation. An equivalent formulation is, given $h \leftarrow\!\!\$\, \mathbb{G}$, find $a \in \mathbb{Z}_p$, $b \in \mathbb{Z}_p^*$ such that $f(g^a/h^b) = b$. Indeed, this immediately yields a CDL solution with $z = a$ and $R = g^a/h^b$.

It is easy to see that if CDL[$\mathbb{G}, f$] holds for some function f, then DL holds in $\mathbb{G}$. On the other hand, note that even for DL-hard $\mathbb{G}$, CDL[$\mathbb{G}, f$] cannot hold for *every* function f. In particular, it is necessary that every non-zero value has a small preimage under f: Suppose $t \in \mathbb{Z}_p^*$ has a large preimage; then a uniform $z \leftarrow\!\!\$\, \mathbb{Z}_p$, together with $R := g^z/h^t$ breaks CDL if $f(R) = t$, which happens with probability proportional to the size of $f^{-1}(t)$. In Section 5, we show that, when modeling $\mathbb{G}$ as an *elliptic-curve generic group* [GS22], this assumption on f is not only necessary but also sufficient for CDL to hold. For an elliptic-curve group, a simple example of a function f satisfying this assumption (as previously argued in [GS22]) is the *ECDSA conversion function* [Nat13], which maps a point $P = (x, y)$ on the curve to its reduced x-coordinate, *i.e.*, $f\colon (x, y) \mapsto x \bmod p$.

<u>TIGHT REDUCTION FROM SCHNORR TO CDL.</u> We give a completely tight reduction in the ROM from EUF-CMA of $\mathsf{Sch}[\mathbb{G}]$ to CDL[$\mathbb{G}$]. Namely, given a forger against $\mathsf{Sch}[\mathbb{G}]$, we construct an adversary solving CDL[$\mathbb{G}, f$] with essentially the same running time and success probability as the assumed forger. The only assumption we make on f is that $|f^{-1}(0)|$, the number of elements f maps to 0, is small.

Given an instance $h \in \mathbb{G}$ of CDL[$\mathbb{G}, f$], the reduction sets h as the public key for the forger, so it does not know the corresponding secret key. When run, the forger makes hash queries and signing queries. Its signing queries are simulated by the CDL adversary as in the standard proof of Schnorr in [PS96]. Namely, on query m_i, the reduction picks $s_i, c_i \leftarrow\!\!\$\, \mathbb{Z}_p$, sets $R_i \leftarrow g^{s_i}/h^{c_i}$, and programs the random oracle at (R_i, m_i) to be c_i, returning (R_i, s_i) as the signature. (Since R_i is uniform, the probability that the RO is already defined at (R_i, m_i) is negligible.)

When answering hash queries, we embed outputs of f into the answers, so we can later use a forgery to break CDL; at the same time, we need to ensure that the returned values are uniformly distributed values in $\mathbb{Z}_p$, independent of the adversary's view. For this, on RO query (R, m), the reduction picks $a, b \leftarrow\!\!\$\, \mathbb{Z}_p$ and programs the RO at (R, m) to $(f(R \cdot h^a \cdot g^b) + a) \bmod p$. We argue that *no matter what f is*, the hash values are independent and uniform.

Now, consider a successful forgery (R, s) on some m, thus

$$g^s = R \cdot h^c \ , \tag{2}$$

where c is the RO response for (R, m). Since m must be different from the queried messages,[4] the RO query (R, m) was made explicitly (either by the adversary

[4] Note that we can actually show *strong* unforgeability, namely that it's even hard for the adversary to forge a new signature on an already-signed message, since $(R, s, m) \neq (R_i, s_i, m_i)$ for all i implies $(R, m) \neq (R_i, m_i)$ for all i, since s_i is uniquely determined by R_i and m_i.

or the game when verifying the forgery), that is, the RO was not programmed during a signing query. Let thus a, b be the values chosen by the reduction when answering this RO query, that is, $c = (f(R \cdot h^a \cdot g^b) + a) \bmod p$. Together with Eq. (2), this yields $g^s = R \cdot h^{f(R \cdot h^a \cdot g^b)+a}$, which implies

$$g^s \cdot g^b = (R \cdot h^a \cdot g^b) \cdot h^{f(R \cdot h^a \cdot g^b)} .$$

Thus, $(R^* := R \cdot h^a \cdot g^b, z^* := (s + b) \bmod p)$ is a solution to CDL as long as $f(R^*) \neq 0$, which by our initial assumption on f holds with high probability.

TIGHT REDUCTION FROM Sparkle+ TO CDL. We next consider an advanced primitive built around Schnorr signatures. Namely, a *threshold signature* (TS) scheme, a concept originating in [Des88, DF90] that distributes trust among several signers. There is a single verification key vk and a group of n signers that each possess a share sk_i of the corresponding signing key sk and a threshold t. If at least t signers get together, they can produce a signature that verifies under vk. In general, the signing process can consist of more than one round, where in round j the i-th signer sends a protocol message $pm_{i,j}$ and updates its state.

TS schemes have exploded in popularity partly due to their applications to blockchains and cryptocurrencies. A signing key securing funds can be distributed and thus protected against compromise (an adversary would have to corrupt t key shares) and loss (of up to $n - t$ shares). Because Schnorr signatures are already widely deployed (*e.g.*, in Bitcoin), the most desirable TS schemes are Schnorr-based ones; that is, where (vk, σ) has the same form as in Schnorr. However, the same concerns about (lack of) tight security pertain to TS schemes for Schnorr as to Schnorr itself. Indeed, work on TS schemes for Schnorr typically does not consider tight security at all. We stress that our goal is to prove tight security of existing or in-use schemes as is, *without* changing the scheme itself. In part, this is because we view it as unlikely that practitioners will change what scheme they use simply for the sake of tight security.

A recent two-round TS scheme for Schnorr is FROST [KG20, CKM21]. (As is typical in the literature, the first round is message-independent, so FROST is usually called partially non-interactive.) In stark contrast to basic Schnorr, even existing *loose* security proofs for FROST rely on the OMDL assumption [BCK+22].

We thus consider the three-round scheme Sparkle+ [CKM23a] that succeeded FROST. (Again, the first round is message-independent.) Sparkle+ is natural and efficient, combining Shamir secret sharing of the secret key in the exponent with an initial round of hash-based commitment to the signers' randomness. (For technical reasons, a signature scheme, which could also be Schnorr, is used to sign each party's view of the protocol in one of the rounds.) Crites, Komlo, and Maller give a reduction in the ROM from the security of Sparkle+ (under static corruptions) to DL [CKM23a, Theorem 1], with a bound analogous to the classical one for basic Schnorr. By carefully adapting their proof, we show a *completely tight* reduction to CDL.

We note that prior to Sparkle+ two somewhat similar three-round TS schemes for Schnorr were proposed, namely SimpleTSig [CKM21] and Sparkle [CKM23b]. Unfortunately, these schemes are not (known to be) secure because there is an issue with "inconsistent protocol views" explained in [BLT+24]. (The issue applies equally well to the cases of static and adaptive corruptions.) We therefore concentrate on analyzing $\mathsf{Sparkle}+$, for which there is at least a loose reduction from the scheme's security under static corruptions to DL in the ROM. We also note that such security for $\mathsf{Sparkle}+$ is not known to reduce to EUF-CMA of basic Schnorr; therefore our result for $\mathsf{Sparkle}+$ does not follow from our result for basic Schnorr discussed above.

Analyzing CDL in the EC-GGM. To gain confidence in a new computational hardness assumption in prime-order groups, it has become standard to analyze it in the generic group model (GGM) [Nec94, Sho97], where the adversary only gets (random) labels of group elements and has access to an oracle to compute the group operation. Concretely, given the labels of two group elements, the oracle returns the label of the product of the group elements.

This model is however syntactically ill-defined when there is a function taking as input group elements, for example the conversion function in ECDSA [Nat13], which we conjecture to be a suitable choice for CDL in such groups. To analyze the security of ECDSA (and variants thereof), Groth and Shoup (GS) [GS22] introduce the *elliptic-curve GGM*, where the labels are random group elements from an elliptic curve (conditioned on preserving simple properties, namely the identity element and inverses).

In Section 5 we analyze $\mathrm{CDL}[\mathbb{G}, f]$ for arbitrary $\mathbb{G}$ and $f\colon \mathbb{G} \to \mathbb{Z}_p$, where $p = |\mathbb{G}|$, in the EC-GGM model, which uses labels from $\mathbb{G}$. We show that CDL holds conditioned on the aforementioned (necessary) property of f: no element in its range can have a large preimage. In particular, we show that the advantage of any adversary is bounded by $((q+1)\cdot \mathsf{MaxSize}_f + 27q^2 + 39q + 15)/(p-1)$, where $\mathsf{MaxSize}_f := \max_{t\in\mathbb{Z}_p}\{|f^{-1}(t)|\}$ is the largest preimage and q is the number of group-oracle queries made by the adversary. In particular, when f is regular (and has a large image) then, in terms of generic security, CDL is maximally hard, and generic security degrades as a function of the largest preimage of f. Note that we use exactly the same property as GS do on the conversion function to prove security of ECDSA in the EC-GGM. As the ECDSA conversion function is 2-to-1, in this case the bound for CDL is essentially the same as the bound for DL [Sho97, GS22]. In other words, CDL in an appropriate elliptic-curve group and for the ECDSA conversion function is about as hard as DL in the EC-GGM.

In sum, we view the ECDSA conversion function as a good candidate for f in the most practically relevant case of an elliptic-curve group $\mathbb{G}$ since it is simple and general (in that it can be defined for any curve) and, most importantly, all output values have small (two-element) preimages, resulting in almost ideal generic security of $\mathrm{CDL}[\mathbb{G}, f]$.

Analyzing CDL in the algebraic bijective ROM. We also consider partially idealizing f. Particularly, we look at how the ECDSA conversion function is modeled in security analyses of ECDSA that idealize the conversion function. However, our analysis is again not tied to the ECDSA conversion function and works for other functions that have similar structure but are more "random" than the ECDSA conversion function. In fact, for such functions, this idealization is even more meaningful than for the ECDSA conversion function. Thus, it seems reasonable to use such an idealized model in our case.

Initial results on ECDSA's security by Brown [Bro02] modeled the conversion function as a RO (in addition to using the GGM), which ignores its obvious structure. To better capture the structure, the *bijective* ROM was proposed [FKP16, FKP17, HK23]. In this model $f = \psi \circ \Pi \circ \varphi$, where φ maps from $\mathbb{G}$ to $\mathbb{A} := \{0,1\}^L$, Π maps from $\mathbb{A}$ to $\mathbb{B} := [2^L - 1]$, and ψ maps from $\mathbb{B}$ to $\mathbb{Z}_p$. Here φ and ψ are standard-model functions, while Π is modeled as a bijective RO.

In fact, CDL for the ECDSA conversion function is a special case of the *semi-logarithm problem* (SLP) introduced by Brown [Bro05] and generalized by Fersch, Kiltz, and Poettering (FKP) [FKP17, Definition 6] with $\rho_0(u,v) = u$ and $\rho_1(u,v) = -v$. FKP show a loose reduction from SDL to DL in the BROM (and hence get a loose reduction for ECDSA). We would like a *tight* reduction in the case of CDL. We manage to do this by additionally assuming that the adversary is *algebraic* wrt. its queries to Π, *i.e.*, we use the algebraic BRO model (ABROM) recently proposed to analyze blind ECDSA [QCY21]. Specifically, we show that the advantage of any CDL adversary in the ABROM is bounded by the hardness of DL in $\mathbb{G}$ plus $((q+1) \cdot \mathsf{MaxSize}_\psi + q^2 + 2q)/p$, where q is the number of queries made by the adversary. Our algebraic assumption on the adversary is important here. If we showed a tight reduction from CDL to DL in the BROM, that would imply (by our main result) a tight reduction from the security of Schnorr signatures to DL in the ROM. Furthermore, our result here does not already follow from [FPS20] because we don't require the CDL adversary to output a representation for its solution.

1.3 Discussion

COMPARISON TO RATIO-BASED TIGHTNESS. While the focus of their work is on the multi-user setting, Kiltz, Masny and Pan (KMP) [KMP16a] give a two-link chain of reductions going from single-user EUF-CMA of Schnorr signatures to DL. The first reduction [KMP16a, Lemma 3.5] goes from passive impersonation of Schnorr's identification protocol to DL, and the second goes from EUF-CMA of Schnorr signatures to the former. While they claim the first reduction is tight, their criterion for tightness is "ratio-based," namely requiring roughly equal *time-to-success ratios* of the assumed adversary and the constructed one. Unfortunately, as discussed by Bellare and Dai (BD) in [BD20, Appendix B], this criterion is problematic and the aforementioned reduction (which, as in [PS96] uses rewinding), has a substantial running-time blowup. As in [BD20], we employ a notion of tightness that requires roughly equal success probabilities and running times *individually*, avoiding such problems. The salient implication is that our results support group sizes for Schnorr that are actually used in practice.

The second reduction [KMP16a, Theorem 1.1] loses a factor q_H even under ratio-based tightness. Overall, as already argued by [BD20], in the single-user setting KMP do not improve on the required size of the underlying group for Schnorr signatures versus classical results.

Relation to instantiability. CDL is related to the problem of instantiating Schnorr signatures (*i.e.*, replacing its RO with a concrete hash function) under a weak security notion called *universal unforgeability under no-message attack* (UUF-NMA) used to show prior *im*possibility results [PV05, GBL08, Seu12, FJS19]. In UUF-NMA, the forgery message is random and given to the adversary, and the adversary gets no signing queries. As compared to instantiating Schnorr signatures under UUF-CMA, CDL differs in that there is no message and f takes solely a group element as input. We also stress that in the above-mentioned impossibility results, UUF-NMA is only considered in the ROM.

Representation-dependence of CDL. Our reduction from EUF-CMA of Schnorr signatures to CDL does not contradict prior impossibility results because the most general of these results [FJS19] only applies to underlying problems that are representation-invariant, *i.e.*, an instance-solution pair remains valid when the representation of the underlying group is changed. However, CDL is representation-dependent. Indeed, Eq. (1) may hold for one group representation but not another, as the value of $f(R)$ depends on the representation.

Falsifiability of CDL. For given $\mathbb{G}$ and f, CDL[$\mathbb{G}, f$] is clearly a falsifiable assumption in the sense of [Nao03]. For our main result, we can also rely on the potentially weaker, *non-falsifiable* CDL[$\mathbb{G}$], requiring that there merely *exists* an f such that CDL[$\mathbb{G}, f$] holds, even if we don't know what it is. This is because we only use the conversion function f *in proofs*, not in real life. (We do not require modification to Schnorr signatures at all.) There have been previous instances of primitives occurring only in proofs, *e.g.* [BM14, GJO16, MOZ22], but to our knowledge it is novel in the context of Schnorr signatures. Note that we will also need that $|f^{-1}(0)|$ for such f is "small." The advantage of the best adversary breaking CDL[$\mathbb{G}, f$] for a given resource usage and the meaning of "small" then determines our bound on the security of Schnorr signatures.

CDL as a stepping-stone. An important problem left open by our work is whether for every group $\mathbb{G}$ of order p (or those of interest) there is a function $f : \mathbb{G} \to \mathbb{Z}_p$ such that there is a reduction from CDL[$\mathbb{G}, f$] to some more-standard assumption in $\mathbb{G}$. For example, constructing f for which there is a tight reduction from CDL[$\mathbb{G}, f$] to DL in $\mathbb{G}$ would, by composition, yield a tight reduction from EUF-CMA of $\mathsf{Sch}[\mathbb{G}]$ to DL in $\mathbb{G}$. Interestingly, this would *not* contradict the above-mentioned impossibility results because our reduction is *non-generic*, since it computes f. Even a loose reduction would be of interest to corroborate CDL. In general, a construction of f could introduce other assumptions. We stress that Schnorr signatures themselves and their instantiation in practice using cryptographic hashing would remain unaffected.

Extensions to Schnorr. We believe CDL can and will be used to justify tight security of Schnorr-based schemes for a host of signature schemes with advanced functionalities, which are of growing importance in cryptographic theory and practice. Above, we discussed our result in this direction for the Schnorr-based threshold signature scheme $\mathsf{Sparkle}+$ [CKM23a]. There are a number of other Schnorr-based threshold signatures schemes one could consider, including [RRJ+22, BLSW24, BLT+24, BDLR24]. Moreover, many recent works build upon Schnorr to achieve signature schemes with other advanced functionalities, such as aggregate signatures [CGKN21], blind signatures [FW24], multisignatures [NRS21], ring signatures [YEL+21], and adaptor signatures [AEE+21]. (We give some representative citations to recent work, not an exhaustive list.) We leave it as future work to extend our results to such primitives.

2 Preliminaries

<u>Notation.</u> If $\boldsymbol{v}$ is a vector then $|\boldsymbol{v}|$ is its length (the number of its coordinates) and v_i is its i-th coordinate. Strings are identified with vectors over $\{0,1\}$, so that $|Z|$ denotes the length of a string Z and Z_i denotes its i-th bit. By ε we denote the empty string or vector. By $x\|y$ we denote the concatenation of strings x, y. If S is a finite set, then $|S|$ denotes its size and we let $x \leftarrow_{\$} S$ denote picking an element of S uniformly at random and assigning it to x. We use $[n]$ to denote the set $\{1,\ldots,n\} \subset \mathbb{N}$. For a set $S \subseteq [n]$, the i-th Lagrange coefficient is defined as $\lambda_i^S := \prod_{j\in S, j\neq i} \frac{j}{j-i}$.

For a function $f\colon A \to B$, we let $\mathsf{Size}_f(b) := |f^{-1}(b)|$ for all $b \in B$ and $\mathsf{MaxSize}_f := \max_{b\in B} \mathsf{Size}_f(b)$. For sets A, B such that $|A| = |B|$, we let $\mathrm{Inj}(A, B)$ denote the set of injections from A to B.

Algorithms may be randomized unless otherwise indicated. If A is an algorithm, we let $y \leftarrow \mathrm{A}^{\mathrm{O}_1,\cdots}(x_1,\ldots;\omega)$ denote running A on inputs $x_1,\ldots$ and coins ω, with oracle access to $\mathrm{O}_1,\ldots$, and assigning the output to y. Moreover, by $y \leftarrow_{\$} \mathrm{A}^{\mathrm{O}_1,\cdots}(x_1,\ldots)$ we denote picking ω uniformly at random and letting $y \leftarrow \mathrm{A}^{\mathrm{O}_1,\cdots}(x_1,\ldots;\omega)$. We let $\mathbf{Out}(\mathrm{A}^{\mathrm{O}_1,\cdots}(x_1,\ldots))$ denote the set of all possible outputs of A when run on inputs $x_1,\ldots$ and with oracle access to $\mathrm{O}_1,\ldots$. Running time is worst-case, which for an algorithm with access to oracles means across all possible replies from the oracles. We use $\perp$ (bot) as a special symbol to denote rejection, and it is assumed not to be in $\{0,1\}^*$.

<u>Games.</u> We use the code-based game-playing framework of BR [BR06]. By $\Pr[\mathrm{G} \Rightarrow y]$ we denote the probability that the execution of game G results in this output being y. In games, integer variables, set variables, boolean variables and string variables are assumed initialized, respectively, to 0, the empty set $\emptyset$, the boolean false and $\perp$.

2.1 Schnorr Signatures and Their Security

<u>Signature schemes.</u> A *signature scheme* with message space MS is a tuple of algorithms $\mathsf{DS} = (\mathsf{DS.K}, \mathsf{DS.S}, \mathsf{DS.V})$ that work as follows:

- $\mathsf{DS.K}$: The key-generation algorithm outputs a key pair (vk, sk). (We suppress the security parameter for simplicity.)
- $\mathsf{DS.S}(sk, m)$: On inputs a signing key sk and a message $m \in \mathsf{MS}$, the signing algorithm outputs a signature σ.

Game $\mathbf{G}_{\mathsf{DS}}^{\text{euf-cma}}$	Game $\mathbf{G}_{\mathsf{DS}}^{\text{mu-euf-cma}}$
INITIALIZE:	INITIALIZE:
$(vk, sk) \leftarrow\$ \mathsf{DS.K}$	$(n, \mathsf{cor}, st) \leftarrow\$ \mathrm{A}_0$ // $\mathsf{cor} \subset [n]$
$S \leftarrow \emptyset$	For $i \in [n]$:
Return vk	$\quad (vk_i, sk_i) \leftarrow\$ \mathsf{DS.K}$; $S_i \leftarrow \emptyset$
	Return $(\{vk_i\}_{i\in[n]}, \{sk_i\}_{i\in\mathsf{cor}}, st)$
SIGNO(m): // $m \in \mathsf{MS}$	SIGNO(i, m): // $i \notin \mathsf{cor}, m \in \mathsf{MS}$
$\sigma \leftarrow\$ \mathsf{DS.S}(sk, m)$	$\sigma \leftarrow\$ \mathsf{DS.S}(sk_i, m)$
$S \leftarrow S \cup \{(m, \sigma)\}$	$S_i \leftarrow S_i \cup \{(m, \sigma)\}$
Return σ	Return σ
FINALIZE(m, σ):	FINALIZE(i, m, σ):
If $(m, \sigma) \in S$ return 0	If $i \in \mathsf{cor} \vee (m, \sigma) \in S_i$ return 0
Return $\mathsf{DS.V}(vk, m, \sigma)$	Return $\mathsf{DS.V}(vk_i, m, \sigma)$

Fig. 1: Games defining EUF-CMA (left) and MU-EUF-CMA (right) of DS.

- $\mathsf{DS.V}(vk, m, \sigma)$: On inputs a verification key vk, signature σ, and message, $m \in \mathsf{MS}$, the verification algorithm outputs a bit.

For correctness, we require that

$$\Pr\left[\mathsf{DS.V}\big(vk, m, \mathsf{DS.S}(sk, m)\big) \Rightarrow 1\right] = 1$$

for all $(sk, vk) \in \mathbf{Out}(\mathsf{DS.K})$ and all $m \in \mathsf{MS}$, where the probability is over the coins for $\mathsf{DS.S}$.

SCHNORR SIGNATURES. Let $\mathbb{G}$ be a cyclic group of prime order $p = |\mathbb{G}|$, generated by g. Let $H\colon \mathbb{G} \times \{0,1\}^* \to \mathbb{Z}_p$ be a hash function. The *Schnorr signature scheme* [FS87, Sch90] $\mathsf{Sch}[\mathbb{G}, H] = (\mathsf{Sch.K}, \mathsf{Sch.S}, \mathsf{Sch.V})$ with message space $\{0,1\}^*$ works as follows. Algorithm $\mathsf{Sch.K}$ chooses $x \leftarrow\$ \mathbb{Z}_p$, sets $X \leftarrow g^x$, and returns $(vk = X, sk = x)$. Algorithm $\mathsf{Sch.S}$ on input x, m chooses $r \leftarrow\$ \mathbb{Z}_p$, sets $R \leftarrow g^r$ and $c \leftarrow H(R, m)$, then returns $\big(R, (r + cx) \bmod p\big)$. Algorithm $\mathsf{Sch.V}$ on inputs $X, m, (R, s)$ returns $\big(g^s = R \cdot X^c\big)$ where $c \leftarrow H(R, m)$. Correctness is straightforward to check. In the ROM, we denote the scheme by $\mathsf{Sch}[\mathbb{G}]$.

EUF-CMA. We define (strong) existential unforgeability under chosen-message attack (EUF-CMA). Let $\mathsf{DS} = (\mathsf{DS.K}, \mathsf{DS.S}, \mathsf{DS.V})$ with message space MS. For an adversary A, we let its EUF-CMA advantage against DS be $\mathbf{Adv}_{\mathsf{DS}}^{\text{euf-cma}}(\mathrm{A}) = \Pr\left[\mathrm{G}_{\mathsf{DS}}^{\text{euf-cma},\mathrm{A}} \Rightarrow 1\right]$, where the game is in Figure 1 (left). We next define multi-user (MU) security. For an adversary $\mathrm{A} = (\mathrm{A}_0, \mathrm{A}_1)$, we let its MU-EUF-CMA advantage against DS be $\mathbf{Adv}_{\mathsf{DS}}^{\text{mu-euf-cma}}(\mathrm{A}) = \Pr\left[\mathrm{G}_{\mathsf{DS}}^{\text{mu-euf-cma},\mathrm{A}} \Rightarrow 1\right]$, where the game is in Figure 1 (right). MU security for signature schemes was introduced by [GMS02]. They proved MU security equivalent to SU security, but with a loose reduction losing at most a factor n in advantage. However, for Schnorr signatures there is a tight reduction from MU to SU security (cf. Remark 4) [KMP16a].

Game $\mathbf{G}^{\mathrm{dl}}_{\mathbb{G},g}$	Game $\mathbf{G}^{\mathrm{cdl}}_{\mathbb{G},g,f}$
INITIALIZE:	INITIALIZE:
$x \leftarrow\!\!\$\, \mathbb{Z}_p$	$x \leftarrow\!\!\$\, \mathbb{Z}_p$
$h \leftarrow g^x$	$h \leftarrow g^x$
Return h	Return h
FINALIZE(x'):	FINALIZE(R, z):
Return $(x = x')$	Return $\big(f(R) \neq 0 \,\wedge\, g^z = R \cdot h^{f(R)}\big)$

Fig. 2: Games defining DL and circular DL problems.

2.2 Discrete-Logarithm Problem

We recall the *discrete-logarithm* (DL) problem. Let $\mathbb{G}$ be a group of prime order $p = |\mathbb{G}|$, generated by g. For an adversary A, we let its DL-advantage against $\mathbb{G}, g$ be $\mathbf{Adv}^{\mathrm{dl}}_{\mathbb{G},g}(\mathrm{A}) = \Pr\left[\mathrm{G}^{\mathrm{dl,A}}_{\mathbb{G},g} \Rightarrow 1\right]$, where the game is in Figure 2.

3 Tight Security of Schnorr Signatures under CDL

We provide our new assumption then proceed to give a tight reduction of EUF-CMA security of Schnorr signatures in the ROM to our assumption.

3.1 Circular Discrete-Logarithm Problem

We introduce the *circular discrete-logarithm* (CDL) problem. Let $\mathbb{G}$ be a group of prime order $p = |\mathbb{G}|$, generated by g. Let $f \colon \mathbb{G} \to \mathbb{Z}_p$. For an adversary A we let its CDL-advantage against $\mathbb{G}, g, f$ be $\mathbf{Adv}^{\mathrm{cdl}}_{\mathbb{G},g,f}(\mathrm{A}) = \Pr\left[\mathbf{G}^{\mathrm{cdl}}_{\mathbb{G},g,f}(\mathrm{A}) \Rightarrow 1\right]$ where the game is in Figure 2.

If we want to assume that there exists no efficient adversary that solves CDL, the condition $f(R) \neq 0$ in the FINALIZE procedure is essential. Otherwise, consider the adversary who has some $R^* \in \mathbb{G}$ such that $f(R^*) = 0$ hard-coded along with $z^* = \mathsf{DLog}_{\mathbb{G},g}(R^*)$, and simply outputs (R^*, z^*) as a valid CDL solution. This adversary would have advantage 1. The assumption would thus be wrong, even though no one might *know* such an adversary. This is analogous to collision-resistance of hash functions, for which an adversary always exists (*cf.* [Rog06]). By adding the condition $f(R) \neq 0$, we simply avoid such issues. The assumption remains strong enough for all our applications.

Remark 1. It is easy to see that CDL being hard for some $\mathbb{G}, f$ such that $|f^{-1}(0)|$ is small implies DL is hard for $\mathbb{G}$. Namely, if there was a DL-adversary A with high advantage then consider the CDL-adversary B that on input h runs A on h to get x such that $g^x = h$, chooses $r \leftarrow\!\!\$\, \mathbb{Z}_p$ at random, and sets $R \leftarrow g^r$. Then B returns $(R, (r + x \cdot f(R)) \bmod p)$, which is a CDL solution if $f(R) \neq 0$. (Note that, strictly speaking, the condition that $|f^{-1}(0)|$ is small is not necessary: Since f is not all-zero, there exists R s.t. $f(R) \neq 0$. If B has R and r with $R = g^r$ hard-coded, it can always compute a valid CDL solution from x.)

Remark 2. Even though CDL is representation-dependent, it can be shown to be random self-reducible. Given an arbitrary CDL instance $h' \in \mathbb{G}$, it can be rerandomized as $h \leftarrow h' \cdot g^{\alpha}$ by sampling $\alpha \leftarrow\!\!\$\, \mathbb{Z}_p$. Then, given a CDL solution (R, z) for h, it is easy to see that $(R, (z - \alpha \cdot f(R)) \bmod p)$ is a CDL solution for the original instance h'.

3.2 Main Result

Theorem 1 *Let $\mathbb{G}$ be a group of prime order p. Let A be an adversary against the Schnorr signature scheme $\mathsf{Sch}[\mathbb{G}]$ in the ROM and assume A makes at most q_s queries to the signing oracle and q_h queries to the hash oracle. Let $f\colon \mathbb{G} \to \mathbb{Z}_p$ be arbitrary and efficient. Then there exists an adversary B with running time roughly the same as A plus simulation overhead proportional to q_s and $q_h \cdot T_f$, where T_f is the time to compute f, such that*

$$\mathbf{Adv}^{\text{euf-cma}}_{\mathsf{Sch}[\mathbb{G}]}(\mathrm{A}) \le \mathbf{Adv}^{\text{cdl}}_{\mathbb{G},g,f}(\mathrm{B}) + \frac{q_s(q_s + q_h) + q_h \cdot \mathsf{Size}_f(0)}{p} \,. \tag{3}$$

Remark 3. The running-time of the adversary constructed in our reduction depends on the running-time for f. Thus, if f is inefficient to compute, it will affect tightness of our reduction. However, all candidate f's we consider in our work are extremely efficient, *e.g.*, they output some bits of the input.

Remark 4. In many implementations of Schnorr signatures, the public key h is prepended to the input to the hash function H. To cover all implementations of Schnorr[5] we do not use such "key-prefixing", but our result readily extends to this setting. Namely, in the reduction, the CDL adversary handles the hash queries (h', R, m) by checking if $h = h'$. If so, it handles the query as in our simulation below. If not, it returns a random value (answering consistently across repeated queries).

Next, we prove the above theorem by formalizing the ideas laid out in the Introduction (Section 1.2).

Proof. We consider a sequence of games defined in Figure 3. Games in boxes (G_1 and G_3) contain the boxed lines, whereas they are ignored for the other games. Note that G_0 is equivalent to $\mathbf{G}^{\text{euf-cma}}_{\mathsf{Sch}[\mathbb{G}]}$.

$\underline{\mathrm{G}_0 \to \mathrm{G}_1.}$ We start with analyzing how the probability of returning 1 changes from G_0 to G_1. The probability of aborting in line 5 in G_1 during any signing query is at most $(q_s + q_h)/p$ since R is uniformly sampled and there are at most $q_s + q_h$ possible $(\cdot, m)$ pairs defined in T it could hit. By a union bound over all signing queries, the probability of G_1 aborting is at most $q_s(q_s + q_h)/p$. Thus,

[5] For example, the standards BSI-TR-03111 (https://www.bsi.bund.de/SharedDocs/Downloads/EN/BSI/Publications/TechGuidelines/TR03111/BSI-TR-03111_V-2-1_pdf.pdf?__blob=publicationFile) and ISO/IEC 14888-3 (https://www.iso.org/standard/76382.html) do not use key-prefixing.

$\mathrm{G}_0(\mathbb{G}, g)$ / $\boxed{\mathrm{G}_1(\mathbb{G}, g)}$	$\mathrm{G}_2(\mathbb{G}, g)$	$\mathrm{G}_3(\mathbb{G}, g, f)$ / $\boxed{\mathrm{G}_4(\mathbb{G}, g, f)}$
INITIALIZE:	INITIALIZE:	INITIALIZE:
1 $S \leftarrow \emptyset$; $\mathsf{T} \leftarrow ()$	1 $S \leftarrow \emptyset$; $\mathsf{T} \leftarrow ()$	1 $S \leftarrow \emptyset$; $\mathsf{T} \leftarrow ()$
2 $x \leftarrow\!\!\$\ \mathbb{Z}_p$; $h \leftarrow g^x$	2 $x \leftarrow\!\!\$\ \mathbb{Z}_p$; $h \leftarrow g^x$	2 $x \leftarrow\!\!\$\ \mathbb{Z}_p$; $h \leftarrow g^x$
3 Return h	3 Return h	3 Return h
SIGNO(m):	SIGNO(m):	SIGNO(m):
4 $r \leftarrow\!\!\$\ \mathbb{Z}_p$; $R \leftarrow g^r$	4 $s \leftarrow\!\!\$\ \mathbb{Z}_p$; $c \leftarrow\!\!\$\ \mathbb{Z}_p$	4 $s \leftarrow\!\!\$\ \mathbb{Z}_p$; $c \leftarrow\!\!\$\ \mathbb{Z}_p$
5 $\boxed{\text{If } \mathsf{T}(R, m) \neq \perp\text{: Abort}}$	5 $R \leftarrow g^s / h^c$	5 $R \leftarrow g^s / h^c$
6 $c \leftarrow$ HASHO(R, m)	6 If $\mathsf{T}(R, m) \neq \perp$: Abort	6 If $\mathsf{T}(R, m) \neq \perp$: Abort
7 $s \leftarrow (r + cx) \bmod p$	7 $\mathsf{T}(R, m) \leftarrow c$	7 $\mathsf{T}(R, m) \leftarrow c$
8 $S \leftarrow S \cup \{(m, R, s)\}$	8 $S \leftarrow S \cup \{(m, R, s)\}$	8 $S \leftarrow S \cup \{(m, R, s)\}$
9 Return (R, s)	9 Return (R, s)	9 Return (R, s)
HASHO(R, m):	HASHO(R, m):	HASHO(R, m):
10 If $\mathsf{T}(R, m) = \perp$:	10 If $\mathsf{T}(R, m) = \perp$:	10 If $\mathsf{T}(R, m) = \perp$:
11 $c \leftarrow\!\!\$\ \mathbb{Z}_p$	11 $c \leftarrow\!\!\$\ \mathbb{Z}_p$	11 $a, b \leftarrow\!\!\$\ \mathbb{Z}_p$
12 $\mathsf{T}(R, m) \leftarrow c$	12 $\mathsf{T}(R, m) \leftarrow c$	12 $R' \leftarrow R \cdot h^a \cdot g^b$
13 Return $\mathsf{T}(R, m)$	13 Return $\mathsf{T}(R, m)$	13 $\boxed{\text{If } f(R') = 0\text{: Abort}}$
		14 $c \leftarrow (f(R') + a) \bmod p$
		15 $\mathsf{T}(R, m) \leftarrow c$
		16 Return $\mathsf{T}(R, m)$
FINALIZE(m, R, s):	FINALIZE(m, R, s):	FINALIZE(m, R, s):
14 If $(m, \sigma) \in S$:	14 If $(m, R, s) \in S$:	17 If $(m, R, s) \in S$:
15 Return 0	15 Return 0	18 Return 0
16 $c \leftarrow$ HASHO(R, m)	16 $c \leftarrow$ HASHO(R, m)	19 $c \leftarrow$ HASHO(R, m)
17 Return $(g^s = Rh^c)$	17 Return $(g^s = Rh^c)$	20 Return $(g^s = Rh^c)$

Fig. 3: Games for the proof of Theorem 1. Changes are highlighted in blue.

by the Fundamental Lemma of Game Playing [BR06] we have $\Pr\left[\mathrm{G}_0^{\mathrm{A}} \Rightarrow 1\right] \leq \Pr\left[\mathrm{G}_1^{\mathrm{A}} \Rightarrow 1\right] + q_s(q_s + q_h)/p$.

$\underline{\mathrm{G}_1 \to \mathrm{G}_2}$. In G_2, we change the signing oracle so that it doesn't require knowledge of the secret key x. We claim these games are equivalent. This is because, if $\mathsf{T}(R, m)$ has not been defined yet, the distribution of (R, s, c) in both games is uniform in $\mathbb{G} \times \mathbb{Z}_p \times \mathbb{Z}_p$ conditioned on $g^s = R \cdot h^c$. In other words, this hop corresponds to a reordering of how these variables are defined. We thus have $\Pr\left[\mathrm{G}_1^{\mathrm{A}} \Rightarrow 1\right] = \Pr\left[\mathrm{G}_2^{\mathrm{A}} \Rightarrow 1\right]$.

$\underline{\mathrm{G}_2 \to \mathrm{G}_3}$. In G_3, we change how we answer hash queries. We argue that the distribution of responses of the hash oracle in G_3 is equivalent to the distribution of responses in G_2. In G_2, the distribution of the value sampled in line 11 is uniform in $\mathbb{Z}_p$. This is also the case in G_3: Since a is uniform in $\mathbb{Z}_p$, so is the

Adversary $\mathrm{B}(h)$

$\mathsf{T} \leftarrow ()$; $i \leftarrow 1$
$(m, R, s) \leftarrow\!\!\$\ \mathrm{A}^{\textsc{SignO},\textsc{HashO}}(\mathbb{G}, g, h)$
$\textsc{HashO}(R, m)$ //ensure that value is defined
Let j be such that $R = R_j$ and $m = m_j$. If such j does not exist, abort
$R^* \leftarrow R \cdot h^{a_j} g^{b_j}$; $s^* \leftarrow (s + b_j) \bmod p$
Return (R^*, s^*)

$\textsc{SignO}(m)$:

$s \leftarrow\!\!\$\ \mathbb{Z}_p; c \leftarrow\!\!\$\ \mathbb{Z}_p; R \leftarrow g^s / h^c$
If $\mathsf{T}(R, m) \neq \bot$: Abort
$\mathsf{T}(R, m) \leftarrow c$
Return (R, s)

$\textsc{HashO}(R_i, m_i)$:

If $\mathsf{T}(R_i, m_i) = \bot$ then
  $a_i, b_i \leftarrow\!\!\$\ \mathbb{Z}_p$
  $R' \leftarrow R_i \cdot h^{a_i} \cdot g^{b_i}$
  If $f(R') = 0$: Abort
  $\mathsf{T}(R_i, m_i) \leftarrow (f(R') + a_i) \bmod p$
  $i \leftarrow i + 1$
Return $\mathsf{T}(R_i, m_i)$

Fig. 4: CDL-adversary B for the proof of Theorem 1.

value $(f(R') + a) \bmod p$. Moreover, conditioned on any *fixed* a, $R' = R \cdot h^a \cdot g^b$ is uniform in $\mathbb{G}$ because b is uniform in $\mathbb{Z}_p$. Thus $\Pr\left[\,\mathrm{G}_2^{\mathrm{A}} \Rightarrow 1\,\right] = \Pr\left[\,\mathrm{G}_3^{\mathrm{A}} \Rightarrow 1\,\right]$.

$\underline{\mathrm{G}_3 \to \mathrm{G}_4}$. This hop introduces an abort whenever a hash query is made and $f(R') = 0$. Since R' is independent and uniformly distributed (as was just shown), the probability of any query aborting is $|f^{-1}(0)|/p$. By a union bound over all queries, the probability of aborting is at most $q_h \cdot |f^{-1}(0)|/p$. Thus, we get $\Pr\left[\,\mathrm{G}_3^{\mathrm{A}} \Rightarrow 1\,\right] \leq \Pr\left[\,\mathrm{G}_4^{\mathrm{A}} \Rightarrow 1\,\right] + q_h \cdot \mathsf{Size}_f(0)/p$.

$\underline{\mathrm{G}_4}$. Combining the hops above yields

$$\mathbf{Adv}^{\text{euf-cma}}_{\mathsf{Sch}[\mathbb{G}]}(\mathrm{A}) \leq \mathbf{Adv}^{\mathrm{G}_4}_{\mathbb{G},g,f}(\mathrm{A}) + \frac{q_s(q_s + q_h) + q_h \cdot \mathsf{Size}_f(0)}{p} \ . \tag{4}$$

Finally, consider adversary B defined in Figure 4. To prove the theorem, it remains to show that

$$\mathbf{Adv}^{\mathrm{G}_4}_{\mathbb{G},g,f}(\mathrm{A}) \leq \mathbf{Adv}^{\mathrm{cdl}}_{\mathbb{G},g,f}(\mathrm{B}) \ ,$$

which combined with Eq. (4) completes the proof. To do so, we show that whenever A returns a strong forgery (m, R, s) in G_4, then B returns a solution to the given CDL instance. A wins if

$$g^s = R \cdot h^c \quad \text{with } c \leftarrow \mathsf{T}(R, m) \ . \tag{5}$$

We claim that $\mathsf{T}(R, m)$ must have been defined during some call to HashO, which in turn means that if A wins, B does not abort in line 4. The reason is that (m, R, s) is different from all (m_i, R_i, s_i), which consist of the i-th query m_i to SignO together with its response (R_i, s_i). Since for a signature (R, s) on m, the value s is determined by R and m, this condition is equivalent to (R, m) being different from all (R_i, m_i). Thus $\mathsf{T}(R, m)$ was not defined during a call to SignO and, if nowhere else, was defined via the call to HashO in line 3.

So let j be the HashO query that defined $\mathsf{T}(R, m)$, and let a_j, b_j be the values that were sampled during the call; thus

$$c = \mathsf{T}(R, m) = \big(f(R \cdot h^{a_j} \cdot g^{b_j}) + a_j\big) \bmod p\,.$$

Together with Eq. (5), this yields $g^s = R \cdot h^{f(R \cdot h^{a_j} \cdot g^{b_j}) + a_j}$. Multiplying by g^{b_j} yields $g^{s+b_j} = (R \cdot h^{a_j} \cdot g^{b_j}) \cdot h^{f(R \cdot h^{a_j} \cdot g^{b_j})}$. We note that $f(R \cdot h^{a_j} \cdot g^{b_j}) \neq 0$ as assured by the abort condition added in G_4. Together this shows that $R^* := R \cdot h^{a_j} \cdot g^{b_j}, s^* := (s + b_j) \bmod p$, the values returned by B, constitute a valid solution to the given CDL instance. $\square$

4 Tight Security of Sparkle+ under CDL

As discussed in the introduction, threshold signature (TS) schemes [Des88, DF90] are an advanced type of signature scheme that distributes trust among n signers, any t of which can collectively create a signature on a common message. Here we treat three-round TS schemes for Schnorr (meaning the final signature produced is a basic Schnorr signature), having an initial round of message-independent preprocessing, following [KG20, CKM21, CKM23a]. In particular, we treat the recent such scheme Sparkle+ [CKM23a].

We restrict our security analysis to static corruptions, meaning the adversary declares which signers it controls at the outset. In this setting, we show a *completely tight* reduction to CDL in the ROM (in addition to multi-user security of an underlying signature scheme, which could also be Schnorr). Comparatively, Crites, Komlo, and Maller [CKM23a] show a loose bound in the ROM under DL, roughly analogous to the classical bound for basic Schnorr.

4.1 The Sparkle+ Scheme and its Security

Our formalizations here largely follow Crites, Komlo, and Maller [CKM23a], with some adaptations to be consistent with our treatment of basic Schnorr and our focus on tight security reductions.

<u>3-ROUND THRESHOLD SIGNATURES.</u> A *three-round threshold signature scheme* with message space MS is a tuple of algorithms

$$\mathsf{TS} = (\mathsf{TS.K}, \{\mathsf{TS.S}_i\}_{i \in \{1,2,3\}}, \mathsf{TS.Agg}, \mathsf{TS.V})$$

that work as follows:

Game $\mathbf{G}^{\mathrm{cor}}_{\mathsf{TS},n,t,\mathcal{S},m}$

$(vk, \{(vk_i, sk_i)\}_{i\in[n]}) \leftarrow\!\!\$\ \mathsf{TS.K}(n,t)$
For $i \in \mathcal{S}$ do:
 $(pm_{i,1}, st_{i,1}) \leftarrow\!\!\$\ \mathsf{TS.S}_1(i, sk_i)$
For $i \in \mathcal{S}$ do:
 $(pm_{i,2}, st_{i,2}) \leftarrow\!\!\$\ \mathsf{TS.S}_2(i, \mathcal{S}, m, \{pm_{j,1}\}_{j\in\mathcal{S}}, st_{i,1})$
For $i \in \mathcal{S}$ do:
 $pm_{i,3} \leftarrow\!\!\$\ \mathsf{TS.S}_3(i, \{pm_{j,2}\}_{j\in\mathcal{S}}, st_{i,2})$
$\sigma \leftarrow \mathsf{TS.Agg}(\{pm_{i,3}\}_{i\in\mathcal{S}})$
Return $\mathsf{TS.V}(vk, m, \sigma)$

Fig. 5: Game defining correctness of TS.

- $\mathsf{TS.K}(n,t)$: The key-generation algorithm, on input the number of signers $n \in \mathbb{N}$ and threshold $t \leq n$, outputs a verification key vk and key shares $\{(vk_i, sk_i)\}_{i\in[n]}$.
- $\mathsf{TS.S}_1(i, sk_i)$: The round-one signing algorithm, on input a signer index $i \in [n]$ and signing key share sk_i, outputs a round-one protocol message $pm_{i,1}$ and signer state $st_{i,1}$.
- $\mathsf{TS.S}_2(i, \mathcal{S}, m, \{pm_{j,1}\}_{j\in\mathcal{S}}, st_{i,1})$: The round-two signing algorithm, on input a signer index $i \in [n]$, set of signers $\mathcal{S} \subseteq [n]$, message $m \in \mathsf{MS}$, round-one protocol messages $\{pm_{j,1}\}_{j\in\mathcal{S}}$, and signer state $st_{i,1}$, outputs a round-two protocol message $pm_{i,2}$ and signer state $st_{i,2}$.
- $\mathsf{TS.S}_3(i, \{pm_{j,2}\}_{j\in\mathcal{S}}, st_{i,2})$: The round-three signing algorithm, on input a signer index $i \in [n]$, round-two protocol messages $\{pm_{j,2}\}_{j\in\mathcal{S}}$, and signer state $st_{i,2}$, outputs a round-three protocol message $pm_{i,3}$.
- $\mathsf{TS.Agg}(\{pm_{j,3}\}_{j\in\mathcal{S}})$: The aggregation algorithm, on input the round-three protocol messages $\{pm_{j,3}\}_{j\in\mathcal{S}}$, outputs a signature σ.
- $\mathsf{TS.V}(vk, m, \sigma)$: The verification algorithm, on input a verification key vk, message $m \in \mathsf{MS}$, and signature σ, outputs a bit.

For correctness, we require that for every $n \in \mathbb{N}$, $t \leq n$, $\mathcal{S} \subseteq [n]$ such that $|\mathcal{S}| \geq t$, and $m \in \mathsf{MS}$, $\Pr\left[\mathbf{G}^{\mathrm{cor}}_{\mathsf{TS},n,t,\mathcal{S},m} \Rightarrow 1\right] = 1$ where the game is in Figure 5.

Note that (non-empty) verification key shares vk_i are not strictly required for all TS schemes, but $\mathsf{Sparkle+}$ has them and they are used to achieve identifiable abort. Moreover, the key-generation algorithm is often implemented via a (verifiable) secret sharing scheme or a distributed key generation (DKG) protocol, as part of which the verification key shares are also revealed.

$\underline{\mathsf{Sparkle+}}$. Let $\mathbb{G}$ be a cyclic group of prime order $p = |\mathbb{G}|$, generated by g. Let $H_{cm} \colon \mathbb{N} \times \mathbb{G} \to \mathbb{Z}_p$ and $H_{sig} \colon \mathbb{G} \times \{0,1\}^* \times \mathbb{G} \to \mathbb{Z}_p$ be hash functions. Let DS be a signature scheme with message-space $\{0,1\}^*$. The $\mathsf{Sparkle+}$ three-round threshold signature scheme [CKM23a]

$$\mathsf{Spar}[\mathbb{G}, H_{cm}, H_{sig}, \mathsf{DS}] = \big(\mathsf{Spar.K}, \{\mathsf{Spar.S}_i\}_{i\in\{1,2,3\}}, \mathsf{Spar.Agg}, \mathsf{Spar.V}\big)$$

with message space $\{0,1\}^*$ is given in Figure 6. In the ROM, we denote the scheme by $\mathsf{Spar}[\mathbb{G}, \mathsf{DS}]$.

To see that $\mathsf{Sparkle}+$ satisfies correctness, we show that the aggregated signature (R, z) satisfies $g^z = R \cdot vk^c$ for $c = H_{sig}(vk, m, R)$. Since $vk = g^x$, and $R = \prod_{i \in \mathcal{S}} R_i$ for $R_i = g^{r_i}$, we can rewrite the verification equation as

$$z = \sum_{i \in \mathcal{S}} r_i + cx \pmod p.$$

Moreover, we have $z = \sum_{i \in \mathcal{S}} z_i$ for $z_i = (r_i + cx_i\lambda_i^{\mathcal{S}}) \bmod p$. Recalling that $\sum_{i \in \mathcal{S}} x_i \lambda_i^{\mathcal{S}} = x$, we get

$$z = \sum_{i \in \mathcal{S}} z_i = \sum_{i \in \mathcal{S}} (r_i + cx_i\lambda_i^{\mathcal{S}}) = \sum_{i \in \mathcal{S}} r_i + c \sum_{i \in \mathcal{S}} x_i \lambda_i^{\mathcal{S}} = \sum_{i \in \mathcal{S}} r_i + cx \pmod p,$$

as required.

TS-EUF-CMA. Our definition of security is adapted from [CKM23a, Definition 7]. We define existential unforgeability under chosen-message attack for TS schemes (TS-EUF-CMA). Let $\mathsf{TS} = (\mathsf{TS.K}, \{\mathsf{TS.S}_i\}_{i \in \{1,2,3\}}, \mathsf{TS.Agg}, \mathsf{TS.V})$ with message space MS be a TS scheme. For an adversary $\mathrm{A} = (\mathrm{A}_0, \mathrm{A}_1)$, we let its TS-EUF-CMA advantage against TS be $\mathbf{Adv}^{\text{ts-euf-cma}}_{\mathsf{TS}}(\mathrm{A}) = \Pr\left[\mathrm{G}^{\text{ts-euf-cma},\mathrm{A}}_{\mathsf{TS}} \Rightarrow 1\right]$, where the game is in Figure 7.

Intuitively, our notion of security allows the adversary to initiate concurrent executions of the signing protocol wherein it controls some fraction of corrupted signers (these corrupted signers being declared at the beginning of the game). Each execution of the signing protocol is given a unique execution identifier *eid*, which represents the fact that protocol messages are tied to a particular execution. (This is important to ensure in any implementation.) The adversary wins if it forges on a message for which it never initiated an execution of the signing protocol.

4.2 Main Result

Crites, Komlo, and Maller give a loose reduction from security of $\mathsf{Sparkle}+$ to DL [CKM23a, Theorem 1]. We show a completely tight reduction under our CDL assumption.

Theorem 2 *Let $\mathbb{G}$ be a group of prime order p. Let DS be a signature scheme. Let A be an adversary against the $\mathsf{Spar}[\mathbb{G}, \mathsf{DS}]$ threshold signature scheme and assume A makes at most q queries to its oracles. Let $f\colon \mathbb{G} \to \mathbb{Z}_p$ be arbitrary and efficient. Then there exist adversaries B_1 and B_2 with running time roughly the same as A plus simulation overhead proportional to $q \cdot T_f$, where T_f is the time to compute f, such that*

$$\mathbf{Adv}^{\text{ts-euf-cma}}_{\mathsf{Spar}[\mathbb{G},\mathsf{DS}]}(\mathrm{A}) \le \mathbf{Adv}^{\text{mu-euf-cma}}_{\mathsf{DS}}(\mathrm{B}_1) + \mathbf{Adv}^{\text{cdl}}_{\mathbb{G},g,f}(\mathrm{B}_2) + \frac{q^2 + q \cdot \mathsf{Size}_f(0)}{p}\,. \tag{6}$$

$\underline{\mathsf{Spar.K}(n,t)\text{:}}$

$x \leftarrow\!\!\$\ \mathbb{Z}_p\ ;\ vk \leftarrow g^x$
$\{(i,x_i)\}_{i\in[n]} \leftarrow\!\!\$\ \mathsf{Share}(n,t,x)$ // Shamir secret sharing
For $i \in [n]$: $X_i \leftarrow g^{x_i}\ ;\ (\mathsf{DS}.vk_i, \mathsf{DS}.sk_i) \leftarrow\!\!\$\ \mathsf{DS.K}\ ;\ vk_i \leftarrow (X_i, \mathsf{DS}.vk_i)$
For $i \in [n]$: $sk_i \leftarrow (x_i, \mathsf{DS}.sk_i, vk, \{vk_j\}_{j\in[n]})$
Return $(vk, \{(vk_i, sk_i)\}_{i\in[n]})$

$\underline{\mathsf{Spar.S}_1(k, sk_k)\text{:}}$

$r_k \leftarrow\!\!\$\ \mathbb{Z}_p\ ;\ R_k \leftarrow g^{r_k}\ ;\ cm_k \leftarrow H_{cm}(k, R_k)$
$st_{k,1} \leftarrow (sk_k, r_k, R_k, cm_k)$
Return $(cm_k, st_{k,1})$

$\underline{\mathsf{Spar.S}_2(k, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}}, st_{k,1})\text{:}}$

$(sk_k, r_k, R_k, cm'_k) \leftarrow st_{k,1}$
If $\mathcal{S} \not\subseteq [n] \vee |\mathcal{S}| < t \vee k \notin \mathcal{S} \vee cm_k \neq cm'_k$: Return $(\perp, \perp)$
$(x_k, \mathsf{DS}.sk_k, vk, \{vk_i\}_{i\in[n]}) \leftarrow sk_k$
$view_k \leftarrow (k, cm_k, R_k, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}})$
$\mathsf{DS}.\sigma_k \leftarrow\!\!\$\ \mathsf{DS.S}(\mathsf{DS}.sk_k, view_k)$
$st_{k,2} \leftarrow (sk_k, r_k, R_k, cm_k, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}}, \mathsf{DS}.\sigma_k)$
Return $((R_k, \mathsf{DS}.\sigma_k), st_{k,2})$

$\underline{\mathsf{Spar.S}_3(k, \{(R_i, \mathsf{DS}.\sigma_i)\}_{i\in\mathcal{S}}, st_{k,2})\text{:}}$

$(sk_k, r_k, R'_k, cm_k, \mathcal{S}', m, \{cm_i\}_{i\in\mathcal{S}'}, \mathsf{DS}.\sigma'_k) \leftarrow st_{k,2}$
If $(R_k, \mathsf{DS}.\sigma_k, \mathcal{S}) \neq (R'_k, \mathsf{DS}.\sigma'_k, \mathcal{S}')$: Return $\perp$
$(x_k, \mathsf{DS}.sk_k, vk, \{(X_i, \mathsf{DS}.vk_i)\}_{i\in[n]}) \leftarrow sk_k$
For $i \in \mathcal{S}$:
  If $cm_i \neq H_{cm}(i, R_i)$: Return $\perp$
  $view_i \leftarrow (i, cm_i, R_i, \mathcal{S}, m, \{cm_j\}_{j\in\mathcal{S}})$
  If $\mathsf{DS.V}(\mathsf{DS}.vk_i, view_i, \mathsf{DS}.\sigma_i) \neq 1$: Return $\perp$
$R \leftarrow \prod_{i\in\mathcal{S}} R_i\ ;\ c \leftarrow H_{sig}(vk, m, R)$
$z_k \leftarrow (r_k + c \cdot x_k \cdot \lambda_k^{\mathcal{S}}) \bmod p$ // $\lambda_k^{\mathcal{S}}$ is the k-th Lagrange coefficient for $\mathcal{S}$
Return (R_k, z_k)

$\underline{\mathsf{Spar.Agg}(\{(R_i, z_i)\}_{i\in\mathcal{S}})\text{:}}$

$R \leftarrow \prod_{i\in\mathcal{S}} R_i\ ;\ z \leftarrow \left(\sum_{i\in\mathcal{S}} z_i\right) \bmod p$
Return (R, z) // same form as basic Schnorr

$\underline{\mathsf{Spar.V}(vk, m, (R, z))\text{:}}$

$c \leftarrow H_{sig}(vk, m, R)$
Return $(g^z = R \cdot vk^c)$

Fig. 6: The $\mathsf{Sparkle+}$ three-round threshold signature scheme.

Proof outline. We formally prove Theorem 2 in Section A; here we give a proof outline.

Let A be an adversary against the TS-EUF-CMA-security of $\mathsf{Spar}[\mathbb{G}, \mathsf{DS}]$. We construct a reduction B_1 against the multi-user security of DS (see Figure 1, right side) and a reduction B_2 against CDL. The adversary B_1 simply simulates

Game $\mathbf{G}^{\text{ts-euf-cma}}_{\mathsf{TS}}$

INITIALIZE:

$\mathsf{Q}_{st}, \mathsf{Q}_m \leftarrow \emptyset$; $eid \leftarrow 0$

$(n, t, \mathsf{cor}, st) \leftarrow\!\!\$\ \mathrm{A}_0$ // $t \leq n, \mathsf{cor} \subset [n], |\mathsf{cor}| < t$

$\mathsf{hon} \leftarrow [n] \setminus \mathsf{cor}$

$(vk, \{(vk_i, sk_i)\}_{i\in[n]}) \leftarrow\!\!\$\ \mathsf{TS.K}(n, t)$

Return $(vk, \{vk_i\}_{i\in[n]}, \{sk_i\}_{i\in\mathsf{cor}}, st)$

SIGNO$_1(k)$: // $k \in \mathsf{hon}$

$eid \leftarrow eid + 1$

$(pm_{k,eid,1}, st_{k,eid,1}) \leftarrow\!\!\$\ \mathsf{TS.S}_1(k, sk_k)$

$\mathsf{Q}_{st}[k, eid, 1] \leftarrow st_{k,eid,1}$

Return $(eid, pm_{k,eid,1})$

SIGNO$_2(k, eid, \mathcal{S}, m, \{pm_{i,eid_i,1}\}_{i\in\mathcal{S}})$: // $k \in \mathsf{hon}, \mathsf{Q}_{st}[k, eid, 1] \neq \perp, \mathsf{Q}_{st}[k, eid, 2] = \perp$

$\mathsf{Q}_m \leftarrow \mathsf{Q}_m \cup \{m\}$

$st_{k,eid,1} \leftarrow \mathsf{Q}_{st}[k, eid, 1]$

$(pm_{k,eid,2}, st_{k,eid,2}) \leftarrow\!\!\$\ \mathsf{TS.S}_2(k, \mathcal{S}, m, \{pm_{i,eid_i,1}\}_{i\in\mathcal{S}}, st_{k,eid,1})$

$\mathsf{Q}_{st}[k, eid, 2] \leftarrow st_{k,eid,2}$

Return $pm_{k,eid,2}$

SIGNO$_3(k, eid, \{pm_{i,eid_i,2}\}_{i\in\mathcal{S}})$: // $k \in \mathsf{hon}, \mathsf{Q}_{st}[k, eid, 2] \neq \perp, \mathsf{Q}_{st}[k, eid, 3] = \perp$

$st_{k,eid,2} \leftarrow \mathsf{Q}_{st}[k, eid, 2]$

$pm_{k,eid,3} \leftarrow\!\!\$\ \mathsf{TS.S}_3(k, \{pm_{2,eid_i,1}\}_{i\in\mathcal{S}}, st_{k,eid,2})$

$\mathsf{Q}_{st}[k, eid, 3] \leftarrow pm_{k,eid,3}$

Return $pm_{k,eid,3}$

FINALIZE(m, σ):

If $m \in \mathsf{Q}_m$: Return 0

Return $\mathsf{TS.V}(vk, m, \sigma)$

Fig. 7: Game defining TS-EUF-CMA of TS.

the game $\mathbf{G}^{\text{ts-euf-cma}}_{\mathsf{Spar}[\mathbb{G},\mathsf{DS}]}$ to A and outputs a forgery if and only if A manages to forge a DS signature on the protocol view of an honest party.

The main idea for reduction B_2 is to use the CDL instance h as the verification key vk together with a simulated Shamir secret sharing of the value $\mathsf{Dlog}_{\mathbb{G},g}(h)$. The generation of the keys for DS is performed honestly. B_2 then simulates the signing protocol for a set of signers $\mathcal{S}$ without knowing the secret keys of the honest parties $\{X_k\}_{k\in\mathsf{hon}}$ as follows:

1. In round 1, it simply outputs random commitments $cm_i \leftarrow\!\!\$\ \mathbb{Z}_p$.
2. In round 2, for every $k \in \mathsf{hon}$, it sets $R_k \leftarrow g^{z_k}/X_k^{c\cdot\lambda_k^{\mathcal{S}}}$ for $z_k \leftarrow\!\!\$\ \mathbb{Z}_p$, and for every $j \in \mathsf{cor}$, it extracts the value R_j from A's queries to H_{cm}. This allows B_2 to compute the corresponding $R \leftarrow \prod_{i\in\mathcal{S}} R_i$ and program the random oracle H_{sig} for a uniformly chosen $c \leftarrow\!\!\$\ \mathbb{Z}_p$ such that $H_{sig}(vk, m, R) = c$. Moreover, it incrementally programs $H_{cm}(k, R_k) \leftarrow cm_k$ for party $k \in \mathsf{hon}$ whenever A makes a query to $\mathsf{Spar.S}_2$ which returns the value R_k.

3. In round 3, it simply outputs the values (R_k, z_k) computed in round 2 for every honest party $k \in \mathsf{hon}$.

If A wins the TS-EUF-CMA game $\mathbf{G}^{\text{ts-euf-cma}}_{\mathsf{Spar}[\mathbb{G},\mathsf{DS}]}$ and does not forge a DS signature, we show that, except with negligible probability, we can extract a CDL solution via the same technique as used in the proof of Theorem 1.

Inconsistency in the original proof. Fix a set of signers $\mathcal{S}$ and a message m. When A calls $\mathsf{Spar.S}_2(k, eid, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}})$ the first time for a party $k \in \mathcal{S}$, Crites *et al.* [CKM23a, Section A, *Case 2 (Valid Query)*, step 4] program the random oracle H_{cm} for all honest parties $i \in \mathcal{S} \cup \mathsf{hon}$. The problem with this approach is that only the commitment cm_k must have come from a previous call to $\mathsf{Spar.S}_1(k, eid)$, where it was chosen uniformly at random, while the other commitments $\{cm_i\}_{i\in(\mathcal{S}\cap\mathsf{hon})\setminus\{k\}}$ are chosen by A. Even though A can't obtain the values $\{R_i\}_{i\in(\mathcal{S}\cap\mathsf{hon})\setminus\{k\}}$ by querying the other honest parties on $\mathsf{Spar.S}_2$, it is able to populate the random oracle H_{cm} with arbitrary values. We fix this inconsistency in the original proof by only gradually programming H_{cm}, keeping the values (i, R_i, cm_i, z_i) in a buffer until explicitly queried by A. We fully specify this in Figure 13, where we highlight the lines dealing with this issue in red.

A corollary. In [KMP16b, Lemma B.1] it is shown that multi-user security of Schnorr has a tight reduction to single-user security. Concretely, for any A that makes at most q_s signing queries there exists B such that

$$\mathbf{Adv}^{\text{mu-euf-cma}}_{\mathsf{Sch}[\mathbb{G}]}(\mathrm{A}) \leq 2 \cdot \mathbf{Adv}^{\text{euf-cma}}_{\mathsf{Sch}[\mathbb{G}]}(\mathrm{B}) + \frac{q_s^2}{p} \; . \tag{7}$$

Using the Schnorr signature scheme $\mathsf{Sch}[\mathbb{G}]$ as DS in $\mathsf{Sparkle+}$ and combining Theorem 1 with Eq. (7), we obtain a completely tight reduction from security of $\mathsf{Sparkle+}$ to CDL, as summarized by the following corollary.

Corollary 3 *Let $\mathbb{G}$ be a group of prime order p. Let A be an adversary against $\mathsf{Spar}[\mathbb{G}, \mathsf{Sch}[\mathbb{G}]]$ and assume A makes at most q queries to its oracles. Let $f\colon \mathbb{G} \to \mathbb{Z}_p$ be arbitrary and efficient. Then there exist an adversary B with running time roughly the same as A plus simulation overhead proportional to $q \cdot T_f$, where T_f is the time to compute f, such that*

$$\mathbf{Adv}^{\text{ts-euf-cma}}_{\mathsf{Spar}[\mathbb{G},\mathsf{Sch}[\mathbb{G}]]}(\mathrm{A}) \leq 3 \cdot \mathbf{Adv}^{\text{cdl}}_{\mathbb{G},g,f}(\mathrm{B}) + \frac{4q^2 + 3q \cdot \mathsf{Size}_f(0)}{p} \; . \tag{8}$$

5 CDL in the Elliptic-Curve GGM

We next turn to justifying CDL in idealized models, showing that for a suitable conversion function, CDL is quantitatively as hard as DL in these models.

The EC-GGM. The *Elliptic-curve generic group model* (EC-GGM) [GS22] was introduced to cover constructions (such as ECDSA) that define a function which takes as input group elements. In contrast to Shoup's [Sho97] original GGM, in the EC-GGM, the "encodings" of the group elements are not random strings but random points on a concrete elliptic curve E, which has a prime number p of points. (Using multiplicative notation, we let 1_E denote the identity element.)

In more detail, in security games defined in the EC-GGM, in the beginning, the challenger chooses a random injective *encoding function* $\tau\colon \mathbb{Z}_p \to E$ which preserves trivial relations; in particular, $\tau(0)$ is the identity element and if $\tau(i) = P$ then $\tau(-i) = P^{-1}$. To "compute" in the group, parties have two oracles: $\textsc{Map}(i)$, for $i \in \mathbb{Z}_p$, returns $\tau(i)$; computing linear combinations of group elements is done by calling $\textsc{Add}(c_1, P_1, c_2, P_2)$ for $c_i \in \mathbb{Z}_p$ and $P_i \in E$, for $i = 1, 2$, which returns $\tau(c_1 \cdot \tau^{-1}(P_1) + c_2 \cdot \tau^{-1}(P_2))$.

Note that for schemes (and assumptions) defined over elliptic curve groups (and in particular, if there is a function whose domain is the group), this model is more realistic than the pure GGM (and makes sense syntactically). *E.g.*, in the GGM, one can show ECDSA strongly unforgeable – although it is malleable – which is not possible in the EC-GGM.

<u>Proof overview of CDL in the EC-GGM.</u> To argue that CDL holds in the EC-GGM, we follow the common approach for GGM proofs, also taken by Groth and Shoup [GS22]: instead of randomly sampling τ in the beginning of the game, we sample entries of the form (i, P) on the fly as required; this does not change the adversary's view. The game is defined in Figure 8.

We then simplify the game and abort whenever the "lazy" sampling does not succeed at the first try (including the boxes with a single frame in Figure 8). Letting q denote the number of the adversary's group oracle queries, the difference to the original game is $\mathcal{O}(q^2/p)$. We next define a *symbolic* game where the secret value x is represented by an indeterminate X and the domain of τ now consists of (linear) polynomials in X (the game including all boxes in Figure 8). Again, using a standard GGM argument that relies on the Schwartz–Zippel lemma, the difference to the previous game is $\mathcal{O}(q^2/p)$. Finally, we argue that the adversary's probability in winning the symbolic game is bounded by $\mathsf{MaxSize}_f/p$.

We note that, compared to Groth and Shoup we avoid the use of asymptotics and give a precise concrete analysis of our assumption in the EC-GGM. Additionally, the proof applies to much more general f than the ECDSA conversion function [Nat13], as we simply require a bound on the largest preimage set. (For the ECDSA conversion function, this is two.)

Theorem 4 *Let E be an elliptic curve of prime order p and $f\colon E \to \mathbb{Z}_p$. Let* A *be an adversary against CDL for f in the EC-GGM with E that makes at most q queries to any of its oracles. Then*

$$\mathbf{Adv}^{\text{ec-ggm-cdl}}_{E,f}(\mathrm{A}) \leq \frac{(q+1)\cdot \mathsf{MaxSize}_f + 27q^2 + 39q + 15}{p-1} .$$

Proof. We proceed via a sequence of games as defined in Figure 8 and analyze their differences.

<u>ec-ggm-cdl→abrt-cdl.</u> First consider the boxed code in lines 8 and 14. Consider the probability that in $\mathbf{G}^{\text{ec-ggm-cdl}}_{E,f}$, during any call to $\textsc{Map}$ or $\textsc{Add}$, a value P or i is sampled (uniformly at random) for which there is already an assignment in

Game $\mathbf{G}^{\text{ec-ggm-cdl}}_{E,f}$; $\boxed{\mathbf{G}^{\text{abrt-cdl}}_{E,f}}$; $\boxed{\boxed{\mathbf{G}^{\text{symb-cdl}}_{E,f}}}$

INITIALIZE:

1 $\tau \leftarrow \{(0, 1_E)\}$
2 $g' \leftarrow \text{MAP}(1)$
3 $\boxed{\boxed{h \leftarrow \text{MAP}(X);\ \text{Return } (g', h)}}$
4 $x \leftarrow\!\!\$\ \mathbb{Z}_p^*; h \leftarrow \text{MAP}(x)$
5 Return (g', h)

MAP(i):

6 If $i \notin \text{Dom}(\tau)$
7 $\quad P \leftarrow\!\!\$\ E^*$
8 $\quad \boxed{\text{if } P \in \text{Ran}(\tau) \text{ then abort}}$
9 $\quad$ while $P \in \text{Ran}(\tau) : P \leftarrow\!\!\$\ E^*$
10 $\quad \tau \leftarrow \tau \cup \{(-i, P^{-1}), (i, P)\}$
11 Return $\tau(i)$

ADD(c_1, P_1, c_2, P_2):

12 For $j = 1, 2$: if $P_j \notin \text{Ran}(\tau)$
13 $\quad i \leftarrow\!\!\$\ \mathbb{Z}_p^*$
14 $\quad \boxed{\text{if } i \in \text{Dom}(\tau) \text{ then abort}}$
15 $\quad$ while $i \in \text{Dom}(\tau) : i \leftarrow\!\!\$\ \mathbb{Z}_p^*$:
16 $\quad \tau \leftarrow \tau \cup \big\{(-i, P_j^{-1}), (i, P_j)\big\}$
17 Return $\text{MAP}\big(c_1 \cdot \tau^{-1}(P_1) + c_2 \cdot \tau^{-1}(P_2)\big)$

FINALIZE(R, z):

18 $\boxed{\text{if } R \notin \text{Ran}(\tau) \text{ or } z \notin \text{Dom}(\tau) \text{ then return } 0}$
19 Return $\big(f(R) \neq 0 \,\wedge\, \text{MAP}(z) = \text{ADD}(1, R, f(R), h)\big)$

Fig. 8: Original and variants of the EC-GGM game for the circular discrete-log problem.

τ (which we call a "collision"). MAP is called once in lines 2 and 4 and at most q times by the adversary, and ADD is called at most q times by the adversary. Each call to MAP samples at most one value of P (for which either P, P^{-1} either both already have an assignment in τ or both do not), and similarly each call to ADD samples at most two values i and possibly a value P when calling MAP in line 17. Now, the probability of a collision in line 2 is zero and in line 4 is $2/(p-1)$. Then, in the worst case the adversary makes q ADD queries. On the first such call, the probability of a collision in line 13 (in either of its possible executions) is at most $4/(p-1) + 6/(p-1)$ and in line 17 is at most $8/(p-1)$. If no collisions happened, then the probability of one happening in the second call is bounded by $10/(p-1) + 12/(p-1) + 14/(p-1)$, and so on.

By a union bound, the overall probability of a collision is thus bounded by

$$\sum_{j=1}^{3q+1} \frac{2j}{p-1} = \frac{(3q+1)(3q+2)}{p-1} = \frac{9q^2 + 9q + 2}{p-1} \ .$$

Finally, consider the boxed code in line 18. The probability that the adversary wins in $\mathbf{G}^{\text{ec-ggm-cdl}}_{E,f}$ although there is no assignment already in τ for one (or both) of its output elements R and z is $1/(p-1)$.

Thus, overall, the difference between the probabilities of the games $\mathbf{G}^{\text{ec-ggm-cdl}}_{E,f}$ and $\mathbf{G}^{\text{abrt-cdl}}_{E,f}$ outputting 1 is bounded by $(9q^2+9q+3)/(p-1)$.

abrt-cdl→symb-cdl. Consider game $\mathbf{G}^{\text{symb-cdl}}_{E,f}$, but where we sample x (in line 4), as in $\mathbf{G}^{\text{abrt-cdl}}_{E,f}$. If we furthermore replace the checks "if $i \in \mathrm{Dom}(\tau)$" (lines 6 and 14), where i is now a linear polynomial in X, by "if $i(x) \in \mathrm{Dom}(\tau)$" (as well as all other occurrences of i by $i(x)$) then we obtain a game that is distributed like the previous game $\mathbf{G}^{\text{abrt-cdl}}_{E,f}$.

The difference between games $\mathbf{G}^{\text{abrt-cdl}}_{E,f}$ and $\mathbf{G}^{\text{symb-cdl}}_{E,f}$ outputting 1 is thus bounded by the probability C that for any two $i \neq i' \in \mathrm{Dom}(\tau)$, we have $i(x) = i'(x)$. Initialize creates 4 polynomials (± 1 and $\pm X$), a call to Add creates up to six polynomials (four in line 16 and two in line 10 when line 17 is executed), while a call to Map creates fewer; and Finalize creates at most 2 different polynomials (since it aborts if one of its arguments is not yet in τ). Note that, as long as no collision occurs, all polynomials are independent of x. Thus, C is upper-bounded by the probability that when sampling a random evaluation point (out of p possible ones) any two out of $6q+6$ different polynomials (which are lines) intersect. Since $6q+6$ lines intersect in at most $\sum_{i=1}^{6q+6}(i-1) = \sum_{i=1}^{6q+5} i = \frac{1}{2}(6q+5)(6q+6) = (6q+5)(3q+3)$ points, we have

$$C = \frac{(3q+2)(6q+6)}{p-1} = \frac{18q^2+30q+12}{p-1} \ .$$

symb-cdl. Consider an output (R, z) by the adversary in game $\mathbf{G}^{\text{symb-cdl}}_{E,f}$ that makes Finalize return 1. Since $R \in \mathrm{Ran}(\tau)$, there exist $a, b \in \mathbb{Z}_p$ s.t.

$$\tau^{-1}(R) = a + bX \ .$$

Moreover, $\textsc{Map}(z) = \textsc{Add}(1, R, f(R), h)\big)$ implies

$$\tau(z) = \tau\big(\tau^{-1}(R) + t \cdot \tau^{-1}(h)\big) \text{ with } \ t := f(R) \neq 0 \ .$$

Since τ is injective and $\tau(X) = h$, this implies

$$\tau^{-1}(R) = z - tX \ ,$$

thus $a = z$ and $b = -t \neq 0$. Consider the point when R is added to the range of τ. Any "fresh" input R to Add will be associated to a constant polynomial in line 16. Since $\tau^{-1}(R)$ is non-constant, R must have been added to τ during a call to Map in line 10. Moreover this call to Map must have been made by the experiment in lines 3 or 17, since the adversary can only call Map on constant polynomials. When Map is called on some fresh $i = a+bX$, the value R is picked uniformly and independently of a and b. The probability that any R satisfies $f(R) = -b$ is thus bounded by $\mathsf{MaxSize}_f/p$. As the adversary can create at most q values this way, and moreover we could have $f(h) = -1$, the probability that the adversary wins game symb-cdl is bounded by $(q+1) \cdot \mathsf{MaxSize}_f/p$.

Adding to this the differences between the previous games yields the bound of the theorem. □

6 CDL in the Algebraic Bijective RO Model

<u>The Algebraic Bijective Random Oracle Model.</u> The *Algebraic Bijective Random Oracle Model* (ABROM) [QCY21] is a combination of the bijective random oracle (BRO) model [FKP16] and the algebraic group model (AGM) [FKL18], originally introduced to analyze blind ECDSA. It idealizes the ECDSA conversion function $f : \mathbb{G}^* \to \mathbb{Z}_p$ in a similar manner as the BRO, by decomposing f into three independent functions $f := \psi \circ \Pi \circ \varphi$, where φ maps from $\mathbb{G}^*$ to $\mathbb{A} := \{0,1\}^L$, Π maps from $\mathbb{A}$ to $\mathbb{B} := [2^L - 1]$, and ψ maps from $\mathbb{B}$ to $\mathbb{Z}_p$. Functions φ and ψ are standard-model (non-idealized), while Π is modeled as a bijective random oracle. The forward direction Π and its inverse Π^{-1} are accessible via the BRO and BRO^{-1} oracles, respectively. The conditions imposed on φ, Π, ψ in our result are meant to ensure f preserves the essential structure of the ECDSA conversion function such as invertibility and being 2-to-1.

Like the AGM, the ABROM keeps track of all "seen" group elements in a vector $\boldsymbol{U}$. We make a slight tweak to the model such that the domain of f and φ is $\mathbb{G}$ instead of $\mathbb{G}^*$, to better represent the CDL conversion function. The BRO oracle now takes as a vector representation $\boldsymbol{p}$ where we let $R = \prod_i U_i^{p_i}$ and outputs some $\beta \in \mathbb{B}$ corresponding to $\Pi(\varphi(R))$. In other words, to call Π on R, the adversary needs to provide a representation for some preimage R of α under φ. This preimage of α under φ is then added to $\boldsymbol{U}$. The BRO^{-1} oracle takes as input some $\beta \in \mathbb{B}$ and outputs some $\alpha \in \mathbb{A}$ corresponding to $\Pi^{-1}(\beta)$. Unlike the AGM, we do *not* require the adversary to give representations for group elements it outputs. Formally, for an adversary A, the CDL game in the ABROM is defined in Figure 9 as G_0. We let A's advantage be $\mathbf{Adv}^{\text{abro-cdl}}_{\mathbb{G},g,\varphi,\psi}(\mathrm{A}) = \Pr[\,\mathrm{G}_0 \Rightarrow 1\,]$.

We recall the following definition from [FKP16] before stating our result.

Definition 5. *(Semi-Injective Function) Let $\mathbb{G}$ be a prime order group p and $\mathbb{A}$ be a set. A function $\varphi : \mathbb{G} \to \mathbb{A}$ is called* semi-injective *if (a) its range $\varphi(\mathbb{G}) \subseteq \mathbb{A}$ is efficiently decidable and (b) it is either injective or 2-to-1 with $\varphi(X) = \varphi(Y)$ always implying $Y \in \{X, X^{-1}\}$.*

Theorem 5 *Let $\mathbb{G}$ be a group of prime order p. Let A be a CDL-adversary in the ABROM making at most q queries to its oracles. Let $\mathbb{A} = \{0,1\}^L$ and $\mathbb{B} = [2^L - 1]$ such that $2^L \geq p$. Let $\varphi \colon \mathbb{G} \to \mathbb{A}$ be semi-injective, and let $\psi \colon \mathbb{B} \to \mathbb{Z}_p$ be arbitrary. Assume ϕ, ψ are efficiently computable. Then there exists a DL-adversary B (shown in Figure 11) with running time roughly the same as A plus simulation overhead proportional to q such that*

$$\mathbf{Adv}^{\text{abro-cdl}}_{\mathbb{G},g,\varphi,\psi}(\mathrm{A}) \leq \mathbf{Adv}^{\text{dl}}_{\mathbb{G},g}(\mathrm{B}) + \frac{(q+1) \cdot \mathsf{MaxSize}_{\psi} + q^2 + 2q}{p} \,. \tag{9}$$

Game $\mathrm{G}_0(\mathbb{G}, g, \varphi, \psi)$

INITIALIZE:

1 $\Pi \leftarrow\$ \mathrm{Inj}(\mathbb{A}, \mathbb{B})$
2 $x \leftarrow\$ \mathbb{Z}_p$
3 $h \leftarrow g^x$
4 $\boldsymbol{U} \leftarrow (g, h)$
5 Return h

BRO($\boldsymbol{p}$):

6 $R \leftarrow \prod_i U_i^{p_i}$
7 $\boldsymbol{U} \leftarrow \boldsymbol{U} \| R \| R^{-1}$
8 $\alpha \leftarrow \varphi(R)$
9 $\beta \leftarrow \Pi(\alpha)$
10 Return β

BRO$^{-1}(\beta)$:

11 $\alpha \leftarrow \Pi^{-1}(\beta)$
12 If $\alpha \in \varphi(\mathbb{G})$:
13 $\quad (R, R^{-1}) \leftarrow \varphi^{-1}(\alpha)$
14 $\quad \boldsymbol{U} \leftarrow \boldsymbol{U} \| R \| R^{-1}$
15 Return α

FINALIZE(R, z):

16 $\alpha \leftarrow \varphi(R)$
17 $\beta \leftarrow \Pi(\alpha)$
18 $c \leftarrow \psi(\beta)$
19 Return $(g^z = Rh^c)$

Game $\mathrm{G}_1(\mathbb{G}, g, \varphi, \psi)$

INITIALIZE:

1 $\Pi \leftarrow \emptyset$
2 $x \leftarrow\$ \mathbb{Z}_p$
3 $h \leftarrow g^x$
4 $\boldsymbol{U} \leftarrow (g, h)$
5 Return h

BRO($\boldsymbol{p}$):

6 $R \leftarrow \prod_i U_i^{p_i}$
7 $\boldsymbol{U} \leftarrow \boldsymbol{U} \| R \| R^{-1}$
8 $\alpha \leftarrow \varphi(R)$
9 If $(\alpha, \cdot) \in \Pi$: Return $\Pi(\alpha)$
10 $\beta \leftarrow\$ \mathbb{B}$
11 If $(\cdot, \beta) \in \Pi$: Abort
12 $\Pi \leftarrow \Pi \cup \{(\alpha, \beta)\}$
13 Return β

BRO$^{-1}(\beta)$:

14 If $(\cdot, \beta) \in \Pi$: Return $\Pi^{-1}(\beta)$
15 $\alpha \leftarrow\$ \mathbb{A}$
16 If $(\alpha, \cdot) \in \Pi$: Abort
17 If $\alpha \in \varphi(\mathbb{G})$:
18 $\quad (R, R^{-1}) \leftarrow \varphi^{-1}(\alpha)$
19 $\quad \boldsymbol{U} \leftarrow \boldsymbol{U} \| R \| R^{-1}$
20 $\Pi \leftarrow \Pi \cup \{(\alpha, \beta)\}$
21 Return α

FINALIZE(R, z):

22 $\alpha \leftarrow \varphi(R)$
23 If $(\alpha, \cdot) \in \Pi$ then $\beta \leftarrow \Pi(\alpha)$
24 Else $\beta \leftarrow\$ \mathbb{B}$; If $(\cdot, \beta) \in \Pi$: Abort
25 $c \leftarrow \psi(\beta)$
26 Return $(g^z = Rh^c)$

Fig. 9: Games G_0, G_1 for the proof of Theorem 5. Changes are highlighted in blue.

Proof. We show the result through a sequence of game hops shown in Figures 9 and 10. and proceed to analyze their differences.

$\mathrm{G}_0 \to \mathrm{G}_1$. In G_1, we lazily sample Π and introduce abort conditions in lines 11, 16, and 24. Note that the oracle responses in G_0 are identically distributed to the oracle responses in G_1 if G_1 does not abort. Thus, $\Pr\left[\mathrm{G}_0^{\mathrm{A}} \Rightarrow 1\right] \leq \Pr\left[\mathrm{G}_1^{\mathrm{A}} \Rightarrow 1\right] + \Pr\left[\mathrm{G}_1^{\mathrm{A}} \text{ aborts}\right]$. We now analyze the probability that G_1 aborts. On the i-th query (to either BRO or BRO^{-1}), the probability of aborting is at

Game $\mathrm{G}_2(\mathbb{G}, g, \varphi, \psi)$

INITIALIZE:

$\mathit{\Pi} \leftarrow \emptyset$
$x \leftarrow\$\ \mathbb{Z}_p$
$h \leftarrow g^x$
$\boldsymbol{U} \leftarrow (g, h)$
Return h

$\mathrm{BRO}(\boldsymbol{p})$:

$R \leftarrow \prod_i U_i^{p_i}$
$\boldsymbol{U} \leftarrow \boldsymbol{U} \| R \| R^{-1}$
$\alpha \leftarrow \varphi(R)$
If $(\alpha, \cdot) \in \mathit{\Pi}$: Return $\mathit{\Pi}(\alpha)$
$\beta \leftarrow\$\ \mathbb{B}$
If $(\cdot, \beta) \in \mathit{\Pi}$: Abort
$\mathit{\Pi} \leftarrow \mathit{\Pi} \cup \{(\alpha, \beta)\}$
Return β

$\mathrm{BRO}^{-1}(\beta)$:

If $(\cdot, \beta) \in \mathit{\Pi}$: Return $\mathit{\Pi}^{-1}(\beta)$
$\alpha \leftarrow\$\ \mathbb{A}$
If $\alpha \in \varphi(\mathbb{G})$:
    $s', t' \leftarrow\$\ \mathbb{Z}_p$
    $R \leftarrow g^{s'} h^{t'}$
    $\alpha \leftarrow \varphi(R)$
    $\boldsymbol{U} \leftarrow \boldsymbol{U} \| R \| R^{-1}$
If $(\alpha, \cdot) \in \mathit{\Pi}$: Abort
$\mathit{\Pi} \leftarrow \mathit{\Pi} \cup \{(\alpha, \beta)\}$
Return α

FINALIZE(R, z):

$\alpha \leftarrow \varphi(R)$
If $(\alpha, \cdot) \in \mathit{\Pi}$ then $\beta \leftarrow \mathit{\Pi}(\alpha)$
Else $\beta \leftarrow\$\ \mathbb{B}$; If $(\cdot, \beta) \in \mathit{\Pi}$: Abort
$c \leftarrow \psi(\beta)$
Return $(g^z = R h^c)$

Game $\mathrm{G}_3/\boxed{\mathrm{G}_4}(\mathbb{G}, g, \varphi, \psi)$

INITIALIZE:

$\mathit{\Pi} \leftarrow \emptyset$
$x \leftarrow\$\ \mathbb{Z}_p$
$h \leftarrow g^x$
$\boldsymbol{U} \leftarrow (g, h)$
$\boldsymbol{s} \leftarrow (1, 0); \boldsymbol{t} \leftarrow (0, 1)$ // $U_i = g^{s_i} h^{t_i}$
Return h

$\mathrm{BRO}(\boldsymbol{p})$:

$R \leftarrow \prod_i U_i^{p_i}$
$\boldsymbol{U} \leftarrow \boldsymbol{U} \| R \| R^{-1}$
$\alpha \leftarrow \varphi(R)$
If $(\alpha, \cdot) \in \mathit{\Pi}$: Return $\mathit{\Pi}(\alpha)$
$s' \leftarrow \langle \boldsymbol{s}, \boldsymbol{p} \rangle$; $t' \leftarrow \langle \boldsymbol{t}, \boldsymbol{p} \rangle$ // $R = g^{s'} h^{t'}$
$\boldsymbol{s} \leftarrow \boldsymbol{s} \parallel s' \parallel -s'$
$\boldsymbol{t} \leftarrow \boldsymbol{t} \parallel t' \parallel -t'$
$\beta \leftarrow\$\ \mathbb{B}$
If $-t' = \psi(\beta)$: Abort
If $(\cdot, \beta) \in \mathit{\Pi}$: Abort
$\mathit{\Pi} \leftarrow \mathit{\Pi} \cup \{(\alpha, \beta)\}$
Return β

$\mathrm{BRO}^{-1}(\beta)$:

If $(\cdot, \beta) \in \mathit{\Pi}$: Return $\mathit{\Pi}^{-1}(\beta)$
$\alpha \leftarrow\$\ \mathbb{A}$
If $\alpha \in \varphi(\mathbb{G})$:
    $s', t' \leftarrow\$\ \mathbb{Z}_p$
    If $-t' = \psi(\beta)$: Abort
    $\boldsymbol{s} \leftarrow \boldsymbol{s} \parallel s' \parallel -s'$
    $\boldsymbol{t} \leftarrow \boldsymbol{t} \parallel t' \parallel -t'$
    $R \leftarrow g^{s'} h^{t'}$
    $\alpha \leftarrow \varphi(R)$
    $\boldsymbol{U} \leftarrow \boldsymbol{U} \| R \| R^{-1}$
If $(\alpha, \cdot) \in \mathit{\Pi}$: Abort
$\mathit{\Pi} \leftarrow \mathit{\Pi} \cup \{(\alpha, \beta)\}$
Return α

FINALIZE(R, z):

$\alpha \leftarrow \varphi(R)$
If $(\alpha, \cdot) \in \mathit{\Pi}$ then $\beta \leftarrow \mathit{\Pi}(\alpha)$
Else $\beta \leftarrow\$\ \mathbb{B}$; If $(\cdot, \beta) \in \mathit{\Pi}$: Abort
$c \leftarrow \psi(\beta)$
If $g^z = R h^c$:
    $\boxed{\text{If } R \notin \boldsymbol{U}\text{: Abort}}$
    Return 1
Else return 0

Fig. 10: Games $\mathrm{G}_2 - \mathrm{G}_4$ for the proof of Theorem 5. Changes are highlighted in blue.

most $(i-1)/2^L$, and the probability of aborting in line 24 is $q/2^L$. Thus,

$$\Pr\left[\,\mathrm{G}_1^{\mathrm{A}} \text{ aborts}\,\right] = \sum_{i=1}^{q} \frac{i}{2^L} = \frac{q(q+1)}{2^{L+1}} \ .$$

<u>$\mathrm{G}_1 \to \mathrm{G}_2$.</u> In G_2, in answering a BRO^{-1} query we resample α if the originally sampled α is in the range of φ, so that this time we learn the representation. We set $R := g^{s'}h^{t'}$ where s' and t' are uniformly sampled from $\mathbb{Z}_p$. Since g and h are generators, this results in a uniformly random R. We then set $\alpha \leftarrow \varphi(R)$. Because φ is either injective or 2-to-1, any $\alpha \in \varphi(\mathbb{G})$ is equally likely to be chosen, so this method of resampling maintains the uniform distribution on α. Thus, $\Pr\left[\,\mathrm{G}_1^{\mathrm{A}} \Rightarrow 1\,\right] = \Pr\left[\,\mathrm{G}_2^{\mathrm{A}} \Rightarrow 1\,\right]$.

<u>$\mathrm{G}_2 \to \mathrm{G}_3$.</u> In G_3, the first change we make is to use vectors $\boldsymbol{s}$ and $\boldsymbol{t}$ to keep track of each "seen" element U_i, so that $U_i = g^{s_i}h^{t_i}$. This change is purely for bookkeeping and does not affect the behavior of the oracles. The second change is that we add abort conditions in lines 15 and 23, which we will use later.

We now analyze the probability of G_3 aborting. On each BRO query, the probability of aborting in a given execution of line 15 is at most $\mathsf{MaxSize}_\psi/p$ since β is uniformly sampled. Thus, by union bound over all the queries, the probability of aborting in line 15 during the execution of the entire game is at most $q \cdot \mathsf{MaxSize}_\psi/p$. The probability of aborting in line 23 is at most q/p since t' is uniformly sampled. Thus,

$$\Pr\left[\,\mathrm{G}_2^{\mathrm{A}} \Rightarrow 1\,\right] \le \Pr\left[\,\mathrm{G}_3^{\mathrm{A}} \Rightarrow 1\,\right] + q/p + q \cdot \mathsf{MaxSize}_\psi/p \ .$$

<u>$\mathrm{G}_3 \to \mathrm{G}_4$.</u> In G_4, we introduce an abort in line 37 if the value R in A's output is not in $\boldsymbol{U}$. We have

$$\Pr\left[\,\mathrm{G}_3^{\mathrm{A}} \Rightarrow 1\,\right] \le \Pr\left[\,\mathrm{G}_4^{\mathrm{A}} \Rightarrow 1\,\right] + \Pr\left[\,\mathrm{G}_4^{\mathrm{A}} \text{ aborts}\,\right] \ .$$

We claim that $\Pr\left[\,\mathrm{G}_4^{\mathrm{A}} \text{ aborts}\,\right] \le \mathsf{MaxSize}_\psi/p$. First, note that if $(\alpha, \cdot) \in \mathit{\Pi}$ in line 33 then the abort in line 37 won't be executed. This is because the BRO and BRO^{-1} procedures always add preimages of α under φ to $\boldsymbol{U}$. Thus, if the game aborts in line 37, the value of α will be "fresh," meaning $(\alpha, \cdot) \notin \mathit{\Pi}$. In this case, $\beta \in \mathbb{B}$ is uniform and independent of A's output (R, z). Thus, over the choice of β, the probability that the if-check in line 37 evaluates to true is at most $\mathsf{MaxSize}_\psi/p$ because there is a unique c satisfying the equation.

<u>The DL-Adversary.</u> Finally, we show that there is a DL-adversary B (given in Figure 11) such that

$$\mathbf{Adv}^{\mathrm{G}_4}_{\mathbb{G},g,\varphi,\psi}(\mathrm{A}) \le \mathbf{Adv}^{\mathrm{dl}}_{\mathbb{G},g}(\mathrm{B}) \ , \tag{10}$$

which combined with the above completes the proof. Namely, we show that on any run of G_4 which outputs 1, B returns the discrete log of h. Note that G_4 outputs 1 iff A returns a valid (R, z) pair and $R \in \boldsymbol{U}$. So, there exists some i such that $R = g^{s_i}h^{t_i}$. Thus, $g^z h^{-f(R)} = g^{s_i}h^{t_i}$. Moreover, $t_i + f(R)$ is non-zero

Adversary $\mathrm{B}(h)$

1 $\mathit{\Pi} \leftarrow \emptyset$
2 $\boldsymbol{U} \leftarrow (g, h)$
3 $\boldsymbol{s} \leftarrow (1, 0); \boldsymbol{t} \leftarrow (0, 1)$ // $U_i = g^{s_i} h^{t_i}$
4 $(R, z) \leftarrow\!\$\; \mathrm{A}^{\mathrm{BRO}, \mathrm{BRO}^{-1}}(\mathbb{G}, g, h)$
5 If $R \notin \boldsymbol{U}$: Return 0
6 Let i such that $R = U_i$
7 $x \leftarrow (z - s_i)/(t_i + f(R))$
8 Return x

$\mathrm{BRO}(R, \boldsymbol{p})$:

9 $R \leftarrow \prod_i U_i^{p_i}$
10 $\boldsymbol{U} \leftarrow \boldsymbol{U} \| R$
11 $\alpha \leftarrow \varphi(R)$
12 If $(\alpha, \cdot) \in \mathit{\Pi}$: Return $\mathit{\Pi}(\alpha)$
13 $s' \leftarrow \langle \boldsymbol{s}, \boldsymbol{p} \rangle$; $t' \leftarrow \langle \boldsymbol{t}, \boldsymbol{p} \rangle$ //$R = g^{s'} h^{t'}$
14 $\boldsymbol{s} \leftarrow \boldsymbol{s} \parallel s' \parallel -s'$
15 $\boldsymbol{t} \leftarrow \boldsymbol{t} \parallel t' \parallel -t'$
16 $\beta \leftarrow\!\$\; \mathbb{B}$
17 If $(\cdot, \beta) \in \mathit{\Pi}$: Abort
18 If $-t' = \psi(\beta)$: Abort
19 $\mathit{\Pi} \leftarrow \mathit{\Pi} \cup \{(\alpha, \beta)\}$
20 Return β

$\mathrm{BRO}^{-1}(\beta)$:

21 If $(\cdot, \beta) \in \mathit{\Pi}$: Return $\mathit{\Pi}^{-1}(\beta)$
22 $\alpha \leftarrow\!\$\; \mathbb{A}$
23 If $\alpha \in \varphi(\mathbb{G})$:
24     $s', t' \leftarrow\!\$\; \mathbb{Z}_p$
25   If $-t' = \psi(\beta)$: Abort
26     $\boldsymbol{s} \leftarrow \boldsymbol{s} \parallel s' \parallel -s'$
27     $\boldsymbol{t} \leftarrow \boldsymbol{t} \parallel t' \parallel -t'$
28     $R \leftarrow g^{s'} h^{t'}$
29     $\alpha \leftarrow \varphi(R)$
30     $\boldsymbol{U} \leftarrow \boldsymbol{U} \| R \| R^{-1}$
31 If $(\alpha, \cdot) \in \mathit{\Pi}$: Abort
32 $\mathit{\Pi} \leftarrow \mathit{\Pi} \cup \{(\alpha, \beta)\}$
33 Return α

Fig. 11: DL-adversary B for the proof of Theorem 5.

as assured by the abort conditions in lines 15 and 23, so $x := (z - s_i)/(t_i + f(R))$ is indeed the discrete log of h. $\square$

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A Security Proof of Sparkle+

We give a proof of Theorem 2 showing that Sparkle+ is tightly secure under CDL in the ROM.

Proof. Let $\mathrm{G}_0 := \mathbf{G}^{\text{ts-euf-cma}}_{\mathsf{Spar}[\mathbb{G},\mathsf{DS}]}$ be the TS-EUF-CMA security game for Sparkle+ in the ROM, given in Figure 12.

$\underline{\mathrm{G}_0 \to \mathrm{G}_1}$. Let G_1 be a modification of G_0 that aborts if the adversary manages to forge a DS signature on a protocol view which was not previously signed by $\textsc{SignO}_2$. Concretely, we add the boxed lines to G_0 in Figure 12.

Let B_1 be the trivial adversary against the multi-user security of the signature scheme DS (Figure 1, right side) that perfectly simulates the game G_1 to the adversary A and wins if and only if the abort in line 26 happens. Then, we have:

$$\Pr\left[\,\mathrm{G}_0^{\mathrm{A}} \Rightarrow 1\,\right] \le \Pr\left[\,\mathrm{G}_1^{\mathrm{A}} \Rightarrow 1\,\right] + \mathbf{Adv}^{\text{mu-euf-cma}}_{\mathsf{DS}}(\mathrm{B}_1) \tag{11}$$

Game $\mathrm{G}_0(\mathbb{G}, g)$ / $\boxed{\mathrm{G}_1(\mathbb{G}, g)}$

$\textsc{Initialize}$:
$\mathsf{Q}_{st}, \mathsf{Q}_m, \mathsf{Q}_{cm}, \mathsf{Q}_{sig}\boxed{, \mathsf{Q}_{\mathsf{DS}}} \leftarrow \emptyset$; $eid \leftarrow 0$
$(n, t, \mathsf{cor}, st) \leftarrow\!\!\$\, \mathrm{A}_0$ // $t \le n, \mathsf{cor} \subset [n], |\mathsf{cor}| < t$
$\mathsf{hon} \leftarrow [n] \setminus \mathsf{cor}$
$(vk, \{(vk_i, sk_i)\}_{i\in[n]}) \leftarrow\!\!\$\, \mathsf{Spar.K}(n, t)$
Return $(vk, \{vk_i\}_{i\in[n]}, \{sk_i\}_{i\in\mathsf{cor}}, st)$

$\textsc{SignO}_1(k)$: // $k \in \mathsf{hon}$
$eid \leftarrow eid + 1$
$r_k \leftarrow\!\!\$\, \mathbb{Z}_p$; $R_k \leftarrow g^{r_k}$; $cm_k \leftarrow \textsc{HashO}_{cm}(k, R_k)$
$\mathsf{Q}_{st}[k, eid, 1] \leftarrow (sk_k, r_k, R_k, cm_k)$
Return (eid, cm_k)

$\textsc{SignO}_2(k, eid, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}})$: // $k \in \mathsf{hon}, \mathsf{Q}_{st}[k, eid, 1] \ne \perp, \mathsf{Q}_{st}[k, eid, 2] = \perp$
$\mathsf{Q}_m \leftarrow \mathsf{Q}_m \cup \{m\}$
$(sk_k, r_k, R_k, cm'_k) \leftarrow \mathsf{Q}_{st}[k, eid, 1]$
If $\mathcal{S} \not\subseteq [n] \vee |\mathcal{S}| < t \vee k \notin \mathcal{S} \vee cm_k \ne cm'_k$: Return $\perp$
$(x_k, \mathsf{DS}.sk_k, vk, \{vk_i\}_{i\in[n]}) \leftarrow sk_k$
$view_k \leftarrow (k, cm_k, R_k, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}})$
$\boxed{\mathsf{Q}_{\mathsf{DS}} \leftarrow \mathsf{Q}_{\mathsf{DS}} \cup \{(k, view_k)\}}$
$\mathsf{DS}.\sigma_k \leftarrow\!\!\$\, \mathsf{DS.S}(\mathsf{DS}.sk_k, view_k)$
$\mathsf{Q}_{st}[k, eid, 2] \leftarrow (sk_k, r_k, R_k, cm_k, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}}, \mathsf{DS}.\sigma_k)$
Return $(R_k, \mathsf{DS}.\sigma_k)$

$\textsc{SignO}_3(k, eid, \{(R_i, \mathsf{DS}.\sigma_i)\}_{i\in\mathcal{S}})$: // $k \in \mathsf{hon}, \mathsf{Q}_{st}[k, eid, 2] \ne \perp, \mathsf{Q}_{st}[k, eid, 3] = \perp$
$(sk_k, r_k, R'_k, cm_k, \mathcal{S}', m, \{cm_i\}_{i\in\mathcal{S}'}, \mathsf{DS}.\sigma'_k) \leftarrow \mathsf{Q}_{st}[k, eid, 2]$
If $(R_k, \mathsf{DS}.\sigma_k, \mathcal{S}) \ne (R'_k, \mathsf{DS}.\sigma'_k, \mathcal{S}')$: Return $\perp$
$(x_k, \mathsf{DS}.sk_k, vk, \{(X_i, \mathsf{DS}.vk_i)\}_{i\in[n]}) \leftarrow sk_k$
For $i \in \mathcal{S}$:
  If $cm_i \ne \textsc{HashO}_{cm}(i, R_i)$: Return $\perp$
  $view_i \leftarrow (i, cm_i, R_i, \mathcal{S}, m, \{cm_j\}_{j\in\mathcal{S}})$
  If $\mathsf{DS.V}(\mathsf{DS}.vk_i, view_i, \mathsf{DS}.\sigma_i) \ne 1$: Return $\perp$
  $\boxed{\text{If } i \in \mathsf{hon} \wedge (i, view_i) \notin \mathsf{Q}_{\mathsf{DS}}\text{: Abort}}$
$R \leftarrow \prod_{i\in\mathcal{S}} R_i$; $c \leftarrow \textsc{HashO}_{sig}(vk, m, R)$
$z_k \leftarrow (r_k + c \cdot x_k \cdot \lambda_k^{\mathcal{S}}) \bmod p$
$\mathsf{Q}_{st}[k, eid, 3] \leftarrow (R_k, z_k)$
Return (R_k, z_k)

$\textsc{HashO}_{cm}(i, R)$:
If $(i, R, cm) \in \mathsf{Q}_{cm}$: Return cm
$cm \leftarrow\!\!\$\, \mathbb{Z}_p$
$\mathsf{Q}_{cm} \leftarrow \mathsf{Q}_{cm} \cup \{(i, R, cm)\}$
Return cm

$\textsc{HashO}_{sig}(X, m, R)$:
If $(X, m, R, c) \in \mathsf{Q}_{sig}$: Return c
$c \leftarrow\!\!\$\, \mathbb{Z}_p$
$\mathsf{Q}_{sig} \leftarrow \mathsf{Q}_{sig} \cup \{(X, m, R, c)\}$
Return c

$\textsc{Finalize}(m, \sigma)$:
If $m \in \mathsf{Q}_m$: Return 0
Return $\mathsf{Spar.V}(vk, m, \sigma)$

Fig. 12: Games G_0 and G_1 for the proof of Theorem 2.

Game $\mathrm{G}_2(\mathbb{G}, g)$

INITIALIZE:

$\mathsf{Q}_{st}, \mathsf{Q}_m, \mathsf{Q}_{cm}, \mathsf{Q}_{sig}, \mathsf{Q}_{\mathsf{DS}}, \mathsf{Q}'_{cm} \leftarrow \emptyset$; ... // the same as in G_1

SIGNO$_1(k)$: // $k \in \mathsf{hon}$

$eid \leftarrow eid + 1$
$cm_k \leftarrow\$ \mathbb{Z}_p$
If $(\cdot, \cdot, cm_k, \cdot) \in \mathsf{Q}_{cm}$: Abort // $\mathsf{BadEvent}_1$
$\mathsf{Q}_{cm} \leftarrow \mathsf{Q}_{cm} \cup \{(\bot, \bot, cm_k, \bot)\}$
$\mathsf{Q}_{st}[k, eid, 1] \leftarrow (sk_k, \bot, \bot, cm_k)$
Return (eid, cm_k)

SIGNO$_2(k, eid, \mathcal{S}, m, \{cm_i\}_{i \in \mathcal{S}})$: // $k \in \mathsf{hon}, \mathsf{Q}_{st}[k, eid, 1] \neq \bot, \mathsf{Q}_{st}[k, eid, 2] = \bot$

$\mathsf{Q}_m \leftarrow \mathsf{Q}_m \cup \{m\}$
$(sk_k, \bot, \bot, cm'_k) \leftarrow \mathsf{Q}_{st}[k, eid, 1]$
If $\mathcal{S} \not\subseteq [n] \vee |\mathcal{S}| < t \vee k \notin \mathcal{S} \vee cm_k \neq cm'_k$: Return $\bot$
$(x_k, \mathsf{DS}.sk_k, vk, \{(X_i, \mathsf{DS}.vk_i)\}_{i \in [n]}) \leftarrow sk_k$
If $(k, \cdot, cm_k, \cdot) \in \mathsf{Q}'_{cm}$:
  $(k, R_k, cm_k, z_k) \leftarrow \mathsf{Q}'_{cm}$
Else if $\left(\bigvee_{i \in \mathcal{S} \cap \mathsf{hon}} (i, \cdot, cm_i, \cdot) \in \mathsf{Q}_{cm}\right) \vee \left(\bigvee_{i \in \mathcal{S} \cap \mathsf{cor}} (i, \cdot, cm_i, \cdot) \notin \mathsf{Q}_{cm}\right)$:
  $R_k \leftarrow\$ \mathbb{G}$; $z_k \leftarrow \bot$ // *Case 1 (Invalid Query)*
Else: // *Case 2 (Valid Query)*
  $c \leftarrow\$ \mathbb{Z}_p$; $R \leftarrow 1_{\mathbb{G}}$
  For $i \in \mathcal{S} \cap \mathsf{hon}$:
    $z_i \leftarrow\$ \mathbb{Z}_p$; $R_i \leftarrow g^{z_i} / X_i^{c \cdot \lambda_i^{\mathcal{S}}}$; $R \leftarrow R \cdot R_i$
    $\mathsf{Q}'_{cm} \leftarrow \mathsf{Q}'_{cm} \cup \{(i, R_i, cm_i, z_i)\}$
  For $i \in \mathcal{S} \cap \mathsf{cor}$:
    $(i, R_i, cm_i, \bot) \leftarrow \mathsf{Q}_{cm}$; $R \leftarrow R \cdot R_i$
  If $(vk, m, R, \cdot) \in \mathsf{Q}_{sig}$: Abort // $\mathsf{BadEvent}_2$
  $\mathsf{Q}_{sig} \leftarrow \mathsf{Q}_{sig} \cup \{(vk, m, R, c)\}$
$\mathsf{Q}_{cm} \leftarrow \mathsf{Q}_{cm} \setminus \{(\bot, \bot, cm_k, \bot)\}$; $\mathsf{Q}_{cm} \leftarrow \mathsf{Q}_{cm} \cup \{(k, R_k, cm_k, z_k)\}$
$view_k \leftarrow (k, cm_k, R_k, \mathcal{S}, m, \{cm_i\}_{i \in \mathcal{S}})$
$\mathsf{Q}_{\mathsf{DS}} \leftarrow \mathsf{Q}_{\mathsf{DS}} \cup \{(k, view_k)\}$
$\mathsf{DS}.\sigma_k \leftarrow\$ \mathsf{DS.S}(\mathsf{DS}.sk_k, view_k)$
$\mathsf{Q}_{st}[k, eid, 2] \leftarrow (sk_k, \bot, R_k, cm_k, \mathcal{S}, m, \{cm_i\}_{i \in \mathcal{S}}, \mathsf{DS}.\sigma_k)$
Return $(R_k, \mathsf{DS}.\sigma_k)$

SIGNO$_3(k, eid, \{(R_i, \mathsf{DS}.\sigma_i)\}_{i \in \mathcal{S}})$: // $k \in \mathsf{hon}, \mathsf{Q}_{st}[k, eid, 2] \neq \bot, \mathsf{Q}_{st}[k, eid, 3] = \bot$

$(sk_k, \bot, R'_k, cm_k, \mathcal{S}', m, \{cm_i\}_{i \in \mathcal{S}'}, \mathsf{DS}.\sigma'_k) \leftarrow \mathsf{Q}_{st}[k, eid, 2]$
If $(R_k, \mathsf{DS}.\sigma_k, \mathcal{S}) \neq (R'_k, \mathsf{DS}.\sigma'_k, \mathcal{S}')$: Return $\bot$
$(x_k, \mathsf{DS}.sk_k, vk, \{(X_i, \mathsf{DS}.vk_i)\}_{i \in [n]}) \leftarrow sk_k$
For $i \in \mathcal{S}$:
  If $cm_i \neq$ HASHO$_{cm}(i, R_i)$: Return $\bot$
  $view_i \leftarrow (i, cm_i, R_i, \mathcal{S}, m, \{cm_j\}_{j \in \mathcal{S}})$
  If $\mathsf{DS.V}(\mathsf{DS}.vk_i, view_i, \mathsf{DS}.\sigma_i) \neq 1$: Return $\bot$
  If $i \in \mathsf{hon} \wedge (i, view_i) \notin \mathsf{Q}_{\mathsf{DS}}$: Abort
$z_k \leftarrow \mathsf{Q}_{cm}[k, R_k, cm_k]$
If $z_k = \bot$: Abort // $\mathsf{BadEvent}_5$
$\mathsf{Q}_{st}[k, eid, 3] \leftarrow (R_k, z_k)$
Return (R_k, z_k)

HASHO$_{cm}(i, R)$:

If $(i, R, cm, \cdot) \in \mathsf{Q}_{cm}$: Return cm
$cm \leftarrow\$ \mathbb{Z}_p$
If $(\cdot, \cdot, cm, \cdot) \in \mathsf{Q}_{cm}$: Abort // $\mathsf{BadEvent}_1$
$\mathsf{Q}_{cm} \leftarrow \mathsf{Q}_{cm} \cup \{(i, R, cm, \bot)\}$
Return cm

Fig. 13: Game G_2 for the proof of Theorem 2. Changes from G_1 are highlighted in blue. Corrections to the proof in [CKM23a] are highlighted in red.

$\underline{\mathrm{G}_1 \to \mathrm{G}_2}$. Let us define game G_2 as the modified version of G_1 given in Figure 13, where added lines are highlighted in blue. Moreover, our corrections to the proof in [CKM23a, Section A] are highlighted in red.

By a *successful* query to SignO_1, SignO_2, or SignO_3, we mean a query that passes all the checks, and thus doesn't result in the output $\perp$. When it comes to making successful queries, the adversary is restricted in the following way:

1. $\text{SignO}_1(k)$: The adversary can successfully query SignO_1 as much as it wants as long as $k \in \mathsf{hon}$.
2. $\text{SignO}_2(k, eid, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}})$: The adversary must have previously successfully queried $\text{SignO}_1(k)$ with output (eid, cm_k) such that $cm_k \in \{cm_i\}_{i\in\mathcal{S}}$.
3. $\text{SignO}_3(k, eid, \{(R_i, \mathsf{DS}.\sigma_i)\}_{i\in\mathcal{S}})$: The adversary must have successfully queried $\text{SignO}_2(k, eid, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}})$ with output $(R_k, \mathsf{DS}.\sigma_k) \in \{(R_i, \mathsf{DS}.\sigma_i)\}_{i\in\mathcal{S}}$. Moreover, for every $j \in \mathcal{S}$ and $view_j := (j, cm_j, R_j, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}})$, it must hold that $cm_j = \text{HashO}_{cm}(j, R_j)$ and $\mathsf{DS.V}(\mathsf{DS}.vk_j, view_j, \mathsf{DS}.\sigma_j) = 1$.
   Note that the game aborts in line 38 if, for some $j \in (\mathcal{S} \setminus \{k\}) \cap \mathsf{hon}$, A hasn't successfully queried $\text{SignO}_2(j, eid_j, \mathcal{S}, m, \{cm_i\}_{i\in\mathcal{S}})$ with output $(R_j, \mathsf{DS}.\sigma_j) \in \{(R_i, \mathsf{DS}.\sigma_i)\}_{i\in\mathcal{S}}$, as this implies it must have forged the signature $\mathsf{DS}.\sigma_j$.

Without loss of generality, we will assume that the adversary A only makes successful queries (as it can efficiently check whether a query would return $\perp$ before making it).

Next, we argue that (from the view of the adversary) G_2 is equivalent to G_1, except for the aborts happening when one of the events $\mathsf{BadEvent}_1$, $\mathsf{BadEvent}_2$, or $\mathsf{BadEvent}_5$ occurs:[6]

1. Initialize is the same.
2. HashO_{sig} is the same, except for the abort caused by $\mathsf{BadEvent}_1$. The programming in line 24 of SignO_2 is consistent with a random oracle since it uses a uniformly random value $c \leftarrow\$\ \mathbb{Z}_p$.
3. HashO_{cm} is the same, except for the abort caused by $\mathsf{BadEvent}_1$. The programming in line 25 of SignO_2 is consistent with a random oracle since, due to the checks in lines 9–10, the value cm_k added to Q_{cm} must come from a previous query to $\text{SignO}_1(k)$, where it was chosen uniformly at random.
4. SignO_1 has the same output distribution, producing independent and uniformly random commitments $cm_k \leftarrow\$\ \mathbb{Z}_p$, except for the abort caused by $\mathsf{BadEvent}_1$.
5. SignO_2 has the same output distribution, except for the abort caused by $\mathsf{BadEvent}_2$. The signature $\mathsf{DS}.\sigma_k$ is computed as in G_1, so we only need to argue that $R_k \in \mathbb{G}$ is distributed independently and uniformly and satisfies $\text{HashO}_{cm}(k, R_k) = cm_k$. Both in line 15 and line 19, R_k is chosen as a uniformly random group element. Moreover, line 25 ensures that $\text{HashO}_{cm}(k, R_k) = cm_k$.

[6] The third abort event is called $\mathsf{BadEvent}_5$ to remain consistent with the proof in [CKM23a].

6. SIGNO_3 has the same output distribution, except for the abort caused by $\mathsf{BadEvent}_5$. To see why z_k has the correct distribution when $\mathsf{BadEvent}_5$ does not occur, note that line 25 of SIGNO_2 is the only place in G_2 where $z_k \neq \perp$ could have been added to Q_{cm}. Moreover, since $z_k \neq \perp$, it must have been chosen as part of the computations carried out in *Case 2 (Valid Query)* of SIGNO_2. In the original game, we have $z_k \leftarrow (r_k + c \cdot x_k \cdot \lambda_k^{\mathcal{S}}) \bmod p$ for $c \leftarrow \text{HASHO}_{sig}(vk, m, R)$ and a uniformly random $r_k \leftarrow\!\!\$\, \mathbb{Z}_p$. In G_2, the only difference is that r_k is implicitly sampled as $r_k \leftarrow (z'_k - c \cdot x_k \cdot \lambda_k^{\mathcal{S}}) \bmod p$ for $c \leftarrow \text{HASHO}_{sig}(vk, m, R)$ and a uniformly random $z'_k \leftarrow\!\!\$\, \mathbb{Z}_p$. Clearly, these two distributions are equivalent.
   Note that the value $(i, R_i, cm_i, \perp) \in \mathsf{Q}_{cm}$ retrieved in line 22 of SIGNO_2 is *uniquely* defined for every $i \in \mathcal{S} \cap \mathsf{cor}$ because the check in line 14 not passing implies that $(i, \cdot, cm_i, \cdot) \in \mathsf{Q}_{cm}$, and there are no collisions among the cm values in Q_{cm} (as otherwise, the game would have aborted due to $\mathsf{BadEvent}_1$).

From the argumentation above, it follows that:

$$\Pr\left[\,\mathrm{G}_1^{\mathrm{A}} \Rightarrow 1\,\right] \leq \Pr\left[\,\mathrm{G}_2^{\mathrm{A}} \Rightarrow 1\,\right] + \sum_{i \in \{1,2,5\}} \Pr\left[\,\mathsf{BadEvent}_i\,\right] \tag{12}$$

<u>$\mathsf{BadEvent}_1$.</u> Since at most q values $cm_1, \ldots, cm_q$ get sampled from $\mathbb{Z}_p$, by the birthday bound, the probability that there is a collision is at most:

$$\Pr\left[\,\mathsf{BadEvent}_1\,\right] \leq \frac{q^2 - q}{2p} \tag{13}$$

<u>$\mathsf{BadEvent}_2$.</u> For the i-th query, the probability that $(vk, m, R, \cdot) \in \mathsf{Q}_{sig}$ in line 23 of SIGNO_2 is at most $(i-1)/p$ since R is a uniformly random group element (at least one contributing R_i is of the form $g^{z_i}/X_i^{c \cdot \lambda_i^{\mathcal{S}}}$ for $z_i \leftarrow\!\!\$\, \mathbb{Z}_p$) and there are at most $i-1$ possible entries $(vk, m, \cdot, \cdot)$ defined in Q_{sig} it could hit. By a union bound over at most q queries, we get:

$$\Pr\left[\,\mathsf{BadEvent}_2\,\right] \leq \frac{q^2 - q}{2p} \tag{14}$$

<u>$\mathsf{BadEvent}_5$.</u> Note that the abort in line 40 of SIGNO_3 can only occur if the adversary previously successfully queried SIGNO_1 and SIGNO_2 for the same party k. By the execution logic of SIGNO_2, this means that Q_{cm} contains an entry of the form (k, R_k, cm_k, z_k). Moreover, $z_k = \perp$ is only possible if the check in line 14 passed such that line 15 has been executed. There are two possible cases: either (a) there is some $i \in \mathcal{S} \cap \mathsf{hon}$ such that $(i, \cdot, cm_i, \cdot) \in \mathsf{Q}_{cm}$, or (b) there is some $i \in \mathcal{S} \cap \mathsf{cor}$ such that $(i, \cdot, cm_i, \cdot) \notin \mathsf{Q}_{cm}$ at the time of the query to SIGNO_2.

We will argue that case (a) leads to an abort in line 38 of SIGNO_3 before possibly reaching the abort due to $\mathsf{BadEvent}_5$ in line 40. Note that we can't have $(k, \cdot, cm_k, \cdot) \in \mathsf{Q}_{cm}$ because the query to $\text{SIGNO}_1(k)$ already added the value $(\perp, \perp, cm_k, \perp)$ to Q_{cm} in line 5, and there can't be any collisions

Game $\mathrm{G}_3(\mathbb{G}, g)$

$\textsc{HashO}_{sig}(X, m, R)$:

If $(X, m, R, c) \in \mathsf{Q}_{sig}$: Return c
$a, b \leftarrow\!\!\$\, \mathbb{Z}_p$
$R' \leftarrow R \cdot \mathrm{vk}^a \cdot g^b$
If $f(R') = 0$: Abort
$c \leftarrow (f(R') + a) \bmod p$
$\mathsf{Q}_{sig} \leftarrow \mathsf{Q}_{sig} \cup \{(X, m, R, c)\}$
Return c

Fig. 14: Game G_3 for the proof of Theorem 2. Changes from G_2 are highlighted in blue.

among the cm values in Q_{cm} (as otherwise, the game would have aborted due to $\mathsf{BadEvent}_1$). Thus, case (a) means that for some $i \in (\mathcal{S} \setminus \{k\}) \cap \mathsf{hon}$ the value $(i, \cdot, cm_i, \cdot)$ has been added to Q_{cm} via a query to $\textsc{HashO}_{cm}$. In turn, this means that A can't have obtained the corresponding signature $\mathsf{DS}.\sigma_i$ on the view $view_i = (i, cm_i, R_i, \mathcal{S}, m, \{cm_j\}_{j \in \mathcal{S}})$ of party i by querying $\textsc{SignO}_2$, leading to an abort in line 38 of $\textsc{SignO}_3$.

Assume we are in case (b). From this point on, the only way for the game to reach the abort in line 40 of $\textsc{SignO}_3$ is if the check $cm_i = \textsc{HashO}_{cm}(i, R_i)$ in line 35 passes. This means that the adversary managed to query $\textsc{HashO}_{cm}$ on (i, R_i) such that it sampled the output cm_i and added $(i, R_i, cm_i, \perp)$ to Q_{cm}. This only happens with probability $1/p$. By a union bound over at most q queries, we get:

$$\Pr\left[\,\mathsf{BadEvent}_5\,\right] \leq \frac{q}{p} \tag{15}$$

Combining Equations (12)–(15), we get:

$$\Pr\left[\,\mathrm{G}_1^{\mathrm{A}} \Rightarrow 1\,\right] \leq \Pr\left[\,\mathrm{G}_2^{\mathrm{A}} \Rightarrow 1\,\right] + \frac{q^2}{p} \tag{16}$$

$\underline{\mathrm{G}_2 \to \mathrm{G}_3.}$ Let us define a modified version of G_2, referred to as G_3, which only changes $\textsc{HashO}_{sig}$ as highlighted in blue in Figure 14.

As argued in the proof of Theorem 1, G_3 is equivalent to G_2, except for the abort happening when $f(R') = 0$. Thus:

$$\Pr\left[\,\mathrm{G}_2^{\mathrm{A}} \Rightarrow 1\,\right] \leq \Pr\left[\,\mathrm{G}_3^{\mathrm{A}} \Rightarrow 1\,\right] + \frac{q \cdot \mathsf{Size}_f(0)}{p} \tag{17}$$

$\underline{\mathrm{G}_3.}$ Altogether, we obtain:

$$\Pr\left[\,\mathrm{G}_0^{\mathrm{A}} \Rightarrow 1\,\right] \leq \mathbf{Adv}_{\mathsf{DS}}^{\text{mu-euf-cma}}(\mathrm{B}_1) + \Pr\left[\,\mathrm{G}_3^{\mathrm{A}} \Rightarrow 1\,\right] + \frac{q^2 + q \cdot \mathsf{Size}_f(0)}{p} \tag{18}$$

Consider the reduction B_2 defined in Figure 15 from the TS-EUF-CMA security of $\mathsf{Sparkle+}$ to the hardness of CDL, perfectly simulating the game G_3 to the adversary A.

Adversary $\mathrm{B}_2(h)$

$\mathsf{Q}_{st}, \mathsf{Q}_m, \mathsf{Q}_{cm}, \mathsf{Q}_{sig}, \mathsf{Q}_{\mathsf{DS}}, \mathsf{Q}'_{cm} \leftarrow \emptyset$; $eid \leftarrow 0$
$(n, t, \mathsf{cor}, st) \leftarrow\!\$\ \mathrm{A}_0$ // $t \le n, \mathsf{cor} \subset [n]$, w.l.o.g., assume $|\mathsf{cor}| = t - 1$
$\mathsf{hon} \leftarrow [n] \setminus \mathsf{cor}$
$vk \leftarrow h$ // simulated Shamir secret sharing of $\mathsf{Dlog}_{\mathbb{G},g}(h)$
For $i \in \mathsf{cor}$: $x_i \leftarrow\!\$\ \mathbb{Z}_p$; $X_i \leftarrow g^{x_i}$
For $i \in \mathsf{hon}$: $x_i \leftarrow \perp$; $X_i \leftarrow \left(vk / \prod_{j \in \mathsf{cor}} X_j^{\lambda_j^{\mathsf{cor} \cup \{i\}}}\right)^{\frac{1}{\lambda_i^{\mathsf{cor} \cup \{i\}}}}$
For $i \in [n]$: $(\mathsf{DS}.vk_i, \mathsf{DS}.sk_i) \leftarrow\!\$\ \mathsf{DS.K}$; $vk_i \leftarrow (X_i, \mathsf{DS}.vk_i)$
For $i \in [n]$: $sk_i \leftarrow (x_i, \mathsf{DS}.sk_i, vk, \{vk_j\}_{j \in [n]})$
$inputs \leftarrow (vk, \{vk_i\}_{i \in [n]}, \{sk_i\}_{i \in \mathsf{cor}}, st)$
$(m^*, (R^*, z^*)) \leftarrow\!\$\ \mathrm{A}_1^{\textsc{SignO}_1, \textsc{SignO}_2, \textsc{SignO}_3, \textsc{HashO}_{cm}, \textsc{HashO}_{sig}}(inputs)$
$\textsc{HashO}_{sig}(vk, m^*, R^*)$ // ensure that value is defined
$(c^*, a^*, b^*) \leftarrow \mathsf{Q}_{sig}[vk, m^*, R^*]$
If $(a^*, b^*) = (\perp, \perp)$: Abort
$R \leftarrow R^* \cdot \mathrm{vk}^{a^*} \cdot g^{b^*}$; $z \leftarrow (z^* + b^*) \bmod p$
Return (R, z)

$\textsc{SignO}_1(k)$: // the same as in G_3
...

$\textsc{SignO}_2(k, eid, \mathcal{S}, m, \{cm_i\}_{i \in \mathcal{S}})$: // the same as in G_3 except for
...
  If $(vk, m, R, \cdot, \cdot, \cdot) \in \mathsf{Q}_{sig}$: Abort // $\mathsf{BadEvent}_2$
  $\mathsf{Q}_{sig} \leftarrow \mathsf{Q}_{sig} \cup \{(vk, m, R, c, \perp, \perp)\}$
...

$\textsc{SignO}_3(k, eid, \{(R_i, \mathsf{DS}.\sigma_i)\}_{i \in \mathcal{S}})$: // the same as in G_3
...

$\textsc{HashO}_{cm}(i, R)$: // the same as in G_3
...

$\textsc{HashO}_{sig}(X, m, R)$:
If $(X, m, R, c, \cdot, \cdot) \in \mathsf{Q}_{sig}$: Return c
$a, b \leftarrow\!\$\ \mathbb{Z}_p$
$R' \leftarrow R \cdot \mathrm{vk}^a \cdot g^b$
If $f(R') = 0$: Abort
$c \leftarrow (f(R') + a) \bmod p$
$\mathsf{Q}_{sig} \leftarrow \mathsf{Q}_{sig} \cup \{(X, m, R, c, a, b)\}$
Return c

Fig. 15: CDL-adversary B_2 for the proof of Theorem 2.

When A wins G_3, it outputs a forgery (m^*, σ^*) such that (a) $m^* \notin \mathsf{Q}_m$ and (b) $\mathsf{Spar.V}(vk, m^*, \sigma^*) = 1$. From (a), it follows that A never queried $\textsc{SignO}_2$ on m^*. From (b), we get that $\sigma^* = (R^*, z^*)$ such that $g^{z^*} = R^* \cdot vk^c$ for $(c, \cdot, \cdot) = \mathsf{Q}_{sig}[vk, m^*, R^*]$. The value c must have been programmed via a call to $\textsc{HashO}_{sig}$, because the only other place would have been inside a query to $\textsc{SignO}_2$ on m^*.

Thus, B_2 can use the respective values a^*, b^* used to program $\textsc{HashO}_{sig}$ on (vk, m^*, R^*) to output the CDL solution $(R, z) := (R^* \cdot vk^{a^*} \cdot g^{b^*}, (z^* + b^*) \bmod p)$. As B_2 perfectly simulates G_3, we have:

$$\Pr\left[\,\mathrm{G}_3^{\mathrm{A}} \Rightarrow 1\,\right] \leq \mathbf{Adv}^{\mathrm{cdl}}_{\mathbb{G},g,f}(\mathrm{B}_2) \tag{19}$$

Together with Eq. (18), this concludes the proof of Theorem 2. $\square$